

Self-gravity versus phase mixing in the fixed-angular-momentum Vlasov–Poisson model

Ten particle-in-cell runs across the linear transition

Working notes

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Abstract

We study whether the self-gravity of a small spherical component can prevent phase mixing from erasing a perturbation. The model is the Vlasov–Poisson system in which all particles share the same angular momentum L_0 , embedded in a fixed isochrone background. Linear theory predicts that a collective mode leaves the band of orbital frequencies, and stops being damped, when the ratio η between the self-gravitational shift of the frequencies and the width of the band exceeds $\eta_c \approx 1.1$ – 1.2 . We test this prediction, and the nonlinear behaviour that linear theory does not describe, with ten particle-in-cell (PIC) runs of 10^4 particles on a lowered Maxwellian (Wilson) equilibrium, plus seven unperturbed reference runs. At low amplitude the PIC runs reproduce linear theory: the frequency and damping rate of a mode inside the band agree to 1 %, the frequency of discrete modes to 10^{-4} , and the persistent response of a sharp vacuum edge is recovered. Near the transition, with $\nu = \omega_b/\gamma = 10$, the evolution departs from linear theory as O’Neil’s criterion predicts: damping stops, an island of trapped particles forms at the edge of the band, its width and bounce period agree with the pendulum estimate to 10 % to 20 %, and the mode amplitude oscillates with that period. With $\nu = 3$ the late-time departure is not converged in the number of particles. A discrete mode lying just below the band loses amplitude at $\varepsilon = 0.1$, which linear theory does not predict; the loss is converged and vanishes as ε decreases. With a mass comparable to that of the background ($a_0 = 0.5$, $\eta = 5.1$) the discrete mode is far enough from the band to survive intact even at $\varepsilon = 1$. Finally, the initial condition of earlier runs, a phase-space blob that is not an equilibrium, stops mixing when its mass exceeds ~ 0.03 : it becomes self-bound. Each run has phase-space figures in (r, p_r) and (Q, J) and a video.

1 Theoretical framework

1.1 The fixed-angular-momentum model

We consider a self-gravitating spherical component of mass a_0 moving in a fixed background, the isochrone potential of unit mass and unit scale,

$$\Phi_{\text{iso}}(r) = -\frac{1}{1 + \sqrt{1 + r^2}},$$

with $G = 1$. Masses are measured in units of the background mass. All particles have the same angular momentum $L_0 = 2$, so the distribution function is

$$f(r, p_r, L, t) = F(r, p_r, t) \delta(L - L_0),$$

and F obeys the one-dimensional Vlasov equation with Hamiltonian

$$H = \frac{1}{2}p_r^2 + \Phi_{\text{iso}}(r) + \Phi_{\text{self}}(r, t) + \frac{L_0^2}{2r^2}.$$

The self-consistent potential Φ_{self} solves the Poisson equation with the density

$$4\pi r^2 \rho(r, t) = 8\pi^2 L_0 \int F(r, p_r, t) dp_r,$$

where $8\pi^2 L_0 dr dp_r$ is the phase-space measure of the reduction. The total mass is $a_0 = 8\pi^2 L_0 \iint F dr dp_r$. We write $\Phi_{\text{ef}} = \Phi_{\text{iso}} + \Phi_{\text{self}} + L_0^2/2r^2$ for the effective potential.

1.2 Angle–action variables

Each orbit is a closed curve in (r, p_r) . Its action $J = \frac{1}{2\pi} \oint p_r dr$ is the enclosed area divided by 2π , and the angle Q increases uniformly from 0 to 2π over one radial period. In a fixed potential $\dot{J} = 0$ and $\dot{Q} = \Omega(J)$, the radial frequency. For the isochrone alone

$$\Omega(J) = (J + c)^{-3}, \quad c = \frac{1}{2}(L_0 + \sqrt{L_0^2 + 4}) = 2.414,$$

so the frequency decreases with the action and the circular orbit ($J = 0$) is the fastest. The angle is measured from pericentre: $Q = 0$ at the minimum radius, $Q = \pi$ at the maximum, and $p_r > 0$ in between. Near the circular orbit $r - r_c \approx -\sqrt{2J/\kappa} \cos Q$ and $p_r \approx \sqrt{2J\kappa} \sin Q$, so J measures the size of the epicycle and Q its phase. With self-gravity the map $(r, p_r) \mapsto (Q, J)$ is computed numerically in the equilibrium potential (Section 2.1). All (Q, J) coordinates shown in this document use that fixed map.

1.3 Equilibria and perturbations

A distribution $F_{\text{eq}}(J)$ that depends on the phase-space coordinates only through the action is stationary. Because J is defined in the total potential, which includes the potential of F_{eq} itself, the equilibrium must be self-consistent; its construction is described in Section 2.1. The equilibria are lowered Maxwellians in the energy,

$$F_{\text{eq}} = A E_\gamma(g, x), \quad x = \frac{E_t - E}{T} \quad (E < E_t), \quad E_\gamma(g, x) = \sum_{n \geq 0} \frac{x^{g+n}}{\Gamma(g+n+1)},$$

and $F_{\text{eq}} = 0$ for $E \geq E_t$. Far from the edge $F_{\text{eq}} \propto e^{-E/T}$; near the cut-off energy E_t it vanishes as $(E_t - E)^g$. The Wilson model has $g = 2$ and the King model $g = 1$. The edge is the orbit of action $J_t = 0.138$, $E_t = E(J_t)$, and the temperature is $T = (E_t - E_c)/W_0$ with $W_0 = 3$, where E_c is the energy of the circular orbit. F_{eq} decreases monotonically with the energy, so it has no unstable radial modes (Antonov). We chose it because its vacuum edge is the object of the theorems of Hadžić et al. (Section 1.6).

The equilibrium occupies the actions $0 \leq J \leq J_t$, the support, and its frequencies fill the band $[\Omega_{\text{min}}, \Omega_{\text{max}}] = [\Omega(J_t), \Omega(0)]$ of width $\Delta\Omega$.

The initial conditions are the equilibrium plus a perturbation in angle,

$$F = F_{\text{eq}}(J) [1 + \varepsilon s(J) \cos Q], \quad s = (J/J_t)^{3/2}.$$

It has zero mean in Q , so it changes neither the mass nor the equilibrium. The factor $s(J)$ makes it smooth at the circular orbit, since $J^{3/2} \cos Q \propto J(r - r_c)$. The term $\cos Q$ adds mass on the pericentre side ($Q \approx 0$) and removes it on the apocentre side: the component breathes radially. Since $s \leq 1$, the condition $F \geq 0$ requires $\varepsilon \leq 1$.

1.4 Phase mixing

Without self-gravity each point of (Q, J) rotates at its own frequency, and the perturbation of Section 1.3 evolves as

$$\delta F(Q, J, t) = \varepsilon F_{\text{eq}} s \cos(Q - \Omega(J)t).$$

The initial pattern is sheared: the stripes of $\cos Q$ tilt and become thinner in J . Any linear observable, such as the potential perturbation $\delta\Phi$ or the harmonic h_1 (Section 2.5), is a superposition of phasors $e^{-i\Omega(J)t}$ that dephase, and it decays by the Riemann–Lebesgue lemma. This is phase mixing. There is no dissipation: the information moves to ever finer scales in J . The $k = 1$ harmonic dephases completely in

$$\tau_1 = \frac{2\pi}{\Delta\Omega}.$$

The late-time form of the decay is set by the edges of the band: an edge at which the weight vanishes as $|J - J_e|^\alpha$ produces a tail $\propto t^{-(\alpha+1)}$.

1.5 Collective modes and the coupling parameter

With self-gravity, δF generates a potential $\delta\Phi$ that acts back on the particles. In angle–action variables the Hamiltonian is $H = E(J) + \delta\Phi(Q, J, t)$, where $E(J)$ is the energy of the equilibrium orbit and $\delta\Phi(Q, J, t) = \delta\Phi(r(Q, J), t)$ is the potential perturbation along it. The equations of motion are $\dot{Q} = \Omega(J) + \partial_J \delta\Phi$ and $\dot{J} = -\partial_Q \delta\Phi$, and the Vlasov equation is $\partial_t F + \dot{Q} \partial_Q F + \dot{J} \partial_J F = 0$. With $F = F_{\text{eq}}(J) + \delta F$, and dropping the products of two perturbations, it becomes

$$\partial_t \delta F + \Omega(J) \partial_Q \delta F - F'_{\text{eq}}(J) \partial_Q \delta\Phi = 0.$$

The second term is the free rotation of Section 1.4. The third is the effect of the field on the equilibrium: $\delta\Phi$ changes the actions at the rate $\dot{J} = -\partial_Q \delta\Phi$, and this moves mass along the gradient $F'_{\text{eq}}(J)$. The potential is linear in δF . Each element of phase space is a shell of mass $8\pi^2 L_0 \delta F dQ dJ$, so

$$\delta\Phi(r, t) = 8\pi^2 L_0 \iint G(r, r(Q', J')) \delta F(Q', J', t) dQ' dJ', \quad G(r, r') = -\frac{1}{\max(r, r')}.$$

Why the solutions have the form $e^{i(Q-\omega t)}$. Write the potential as $\delta\Phi = \mathcal{G} \delta F$, where $\mathcal{G}$ is the linear operator defined by the integral above, evaluated at $r = r(Q, J)$. The linearized equation is then

$$i \partial_t \delta F = \mathcal{L} \delta F, \quad \mathcal{L} a = -i [\Omega(J) \partial_Q a - F'_{\text{eq}}(J) \partial_Q (\mathcal{G} a)],$$

where $\mathcal{L}$, the linearized Vlasov–Poisson operator, acts on functions $a(Q, J)$ of phase space and generates the linear evolution. $\mathcal{L}$ does not depend on time, because the equilibrium is stationary. A linear equation with a time-independent operator has the solution $\delta F = a(Q, J) e^{-i\omega t}$ whenever $\mathcal{L} a = \omega a$: ω is an eigenvalue of $\mathcal{L}$ and a its eigenfunction. The sign in $e^{-i\omega t}$ is a convention, the usual one in plasma physics. With it, a pattern $e^{i(Q-\omega t)}$ with $\omega > 0$ advances in Q as the particles do, a free orbit has $\omega = k\Omega(J) > 0$ for $k > 0$, and a complex frequency $\omega - i\gamma$ with $\gamma > 0$ is damped, since $e^{-i(\omega-i\gamma)t} = e^{-i\omega t} e^{-\gamma t}$. The coefficients $\Omega(J)$ and $F'_{\text{eq}}(J)$ do not depend on Q , so we expand in Fourier series, $a = \sum_k a_k(J) e^{ikQ}$ and $\delta\Phi = e^{-i\omega t} \sum_k \delta\Phi_k(J) e^{ikQ}$. Each harmonic then obeys

$$(k\Omega(J) - \omega) a_k(J) = k F'_{\text{eq}}(J) \delta\Phi_k(J).$$

Unlike in a homogeneous plasma, the harmonics do not separate. A shift of Q is not a symmetry, because $\delta\Phi$ depends on Q through the radius $r(Q, J)$, so each $\delta\Phi_k$ depends on all the $a_{k'}$. These relations, with $\delta\Phi_k$ given by the Poisson equation, are the eigenvalue problem $\mathcal{L} a = \omega a$ written in harmonics. Eliminating the a_k leaves a linear equation for $\delta\Phi$ alone, and the condition for it to have a non-zero solution is the dispersion relation. Its roots are the eigenvalues ω . The coupling acts mainly on $k = \pm 1$. Near the circular orbit $r - r_c \approx -\sqrt{2J/\kappa} \cos Q$, with $\kappa = \Omega(0)$ (Section 1.2), so a smooth $\delta\Phi(r)$ seen along an orbit is $\delta\Phi(r_c) - \delta\Phi'(r_c) \sqrt{2J/\kappa} \cos Q$: a constant plus a pure $k = \pm 1$ term. The other harmonics enter at higher order in the size of the epicycle.

The modes are therefore dominated by $k = \pm 1$, and since δF is real the $k = -1$ part is the complex conjugate of the $k = 1$ part, $\propto e^{-i(Q-\omega t)}$, and carries $e^{+i\omega t}$. This is what $\delta F \propto e^{i(Q-\omega t)}$ means. In the PIC runs the $k = 2$ harmonic of the discrete modes is 4–5 % of the $k = 1$ harmonic in D3, 6 % in D4 and 11–12 % in D9 and D10. It is measured with h_k , the h_1 of Section 2.5 with e^{-ikQ_j} .

Discrete modes and Landau damping. Without self-gravity ($\mathcal{G} = 0$), $\mathcal{L}$ multiplies the harmonic k by $k\Omega(J)$. Its spectrum is the continuum of these values, the bands $k [\Omega_{\min}, \Omega_{\max}]$ with $k \neq 0$, and its eigenfunctions are concentrated on single orbits, $a_k \propto \delta(J - J_r)$ with $k\Omega(J_r) = \omega$: each orbit oscillates at its own frequency (Section 1.4). Self-gravity keeps this continuum and can add eigenvalues outside it. With self-gravity, $a_1(J) = F'_{\text{eq}}(J) \delta\Phi_1(J) / (\Omega(J) - \omega)$, where $\delta\Phi_1$ is produced by a_1 itself. All the orbits then oscillate at one frequency, set by the self-consistent field and not by any single orbit, so the solution is collective. Its denominator vanishes at the resonant action, $\Omega(J_r) = \omega$. If ω lies below the band, $0 < \omega < \Omega_{\min}$, no orbit is resonant for any k , because $k\Omega(J) > \omega$ for $k \geq 1$ and $k\Omega(J) < 0$ for $k \leq -1$. If the dispersion relation has a root there, ω is a real eigenvalue of $\mathcal{L}$ with a regular eigenfunction. This is a discrete mode, which oscillates without damping, $\gamma = 0$. If ω lies inside the band, there is no regular solution with real ω . The initial-value problem, solved with a Laplace transform in time, still gives a response whose long-time part oscillates at one frequency and decays, $\propto e^{-i(\omega - i\gamma)t}$. Its complex frequency is not an eigenvalue of $\mathcal{L}$. It is a root of the dispersion relation continued analytically from $\text{Im } \omega > 0$ to $\text{Im } \omega < 0$ (Landau's prescription), and $\gamma > 0$ reflects the exchange of energy with the resonant particles, $\Omega(J) \approx \omega$. This is Landau damping. In both cases $\omega - i\gamma$ is the dominant pole that the fits of Section 2.5 measure, in the PIC runs and in the linear solver alike.

We measure the position of the mode relative to the band by $x = (\omega - \Omega_{\min})/\Delta\Omega$, so that $0 < x < 1$ inside the band and $x < 0$ below it. Gravity is attractive and pushes the mode down, towards $\Omega_{\min}$, the frequency of the largest orbits.

Whether a mode can leave the band depends on two frequency scales. The width of the band, $\Delta\Omega$, sets how fast the perturbation mixes (Section 1.4). The coupling has to be compared with it in the same units, and self-gravity provides a frequency directly. For a nearly circular orbit $\kappa^2 = \Phi'' + 3\Phi'/r_c$, since $\Phi'(r_c) = L_0^2/r_c^3$, and the Poisson equation, $\Phi'' + 2\Phi'/r = 4\pi G\rho$, turns this into

$$\kappa^2 = 4\pi G\rho(r_c) + \frac{GM(< r_c)}{r_c^3}.$$

The mass of the component raises the orbital frequencies by an amount set by its own density at the orbit. The coupling of δF to itself through $\delta\Phi$ comes from the same density and is of the same order. The shift of the frequencies of the equilibrium therefore measures the strength of the coupling, and it needs only the equilibrium. This is a heuristic, not a theorem. We compare the shift of the mean frequency with the width of the band,

$$\eta = \frac{\delta\Omega}{\Delta\Omega/\bar{\Omega}}, \quad \delta\Omega = \frac{\bar{\Omega} - \bar{\Omega}_{\text{iso}}}{\bar{\Omega}_{\text{iso}}},$$

where $\bar{\Omega}$ is the mass-weighted mean frequency of the self-consistent equilibrium and $\bar{\Omega}_{\text{iso}}$ is the same average for the same distribution in the isochrone alone. Since $\bar{\Omega}/\bar{\Omega}_{\text{iso}} = 1 + \delta\Omega$, this is $\eta = (1 + \delta\Omega)(\bar{\Omega} - \bar{\Omega}_{\text{iso}})/\Delta\Omega$. To first order in the shift, η is the number of band widths by which self-gravity moves the mean frequency. In A4 the mean frequency moves by 0.84 band widths and $\eta = 1.00$. In M5, where $\delta\Omega = 1.49$, it moves by 2.05 band widths and $\eta = 5.11$. In a plasma the same role is played by $1/(k\lambda_D) = \omega_p/(kv_{\text{th}})$, the collective frequency over the spread of the free-streaming frequencies. Unlike the mass a_0 alone, η takes the width of the band into account: a narrow band needs less mass than a wide one for the same effect. For $\eta \ll 1$ the frequency spread dominates and the perturbation mixes; for $\eta \gtrsim 1$ the coupling can pull a mode out of the band. The value of η at which the mode reaches the edge of the band, η_c , depends on the family of equilibria, through the width and the shape of the distribution; Section 3.1 gives it for ours.

1.6 The vacuum edge

The distribution vanishes at the edge of the support, $J = J_t$. Hadžić, Rein, Schrecker and Straub (Arch. Ration. Mech. Anal. **249**, 45, 2025) study this same reduction: all particles with the same $|L|$, in the fixed field of a point mass (the isochrone plays this role here), with small equilibria $F = \varepsilon(E_0 - E)^k$. For the linearized problem they prove a dichotomy in the exponent of the edge, valid for ε below a threshold $\varepsilon_0(k)$ that they do not estimate. For $k > 1$ (our g) perturbations are damped. For $1/2 < k \leq 1$ they are not, because there is a discrete mode below $\Omega_{\min}$. Only the behaviour at the edge matters; for the King model ($g = 1$) they note that there is no damping.

A heuristic argument explains why. The dispersion relation contains $\int F'_{\text{eq}}/(\Omega - \omega) dJ$, and with $F'_{\text{eq}} \propto (J_t - J)^{g-1}$ this integral diverges as $\omega \rightarrow \Omega_{\min}$ if $g \leq 1$. An arbitrarily weak coupling then suffices to bind a state at the edge.

1.7 Nonlinear evolution

In a fixed potential the Vlasov equation is exactly linear in F for any amplitude ε ; nonlinearity enters only through $\delta\Phi$. Near a resonance $\Omega(J_r) = \omega$ a particle moves as in a pendulum, with bounce frequency

$$\omega_b = \sqrt{|\Omega'(J_r)| |\delta\Phi|} \propto \sqrt{\varepsilon}.$$

The wave traps particles on a time $\sim 2\pi/\omega_b$ and is damped on a time $1/\gamma$. Their ratio is O'Neil's parameter,

$$\nu = \omega_b/\gamma.$$

For $\nu \ll 1$ the wave is damped before it can trap particles, and linear theory holds at all times. For $\nu \gtrsim 1$ the resonant particles are trapped, damping stops, and the amplitude oscillates and saturates; linear theory does not give the final state. Near the transition $\gamma \rightarrow 0$, so $\nu \rightarrow \infty$ for any amplitude: at $\eta = 1$, $\nu \approx 11\sqrt{\varepsilon}$, and $\varepsilon \approx 0.008$ already gives $\nu = 1$. Whether the evolution is linear is decided by ν , not by ε .

For a discrete mode ($\eta > \eta_c$) there is no resonance inside the band, and the nonlinear question is whether the mode survives at finite amplitude. Its frequency shifts with the amplitude, and a mode just below the edge could enter the band. The relevant ratio compares the bounce frequency with the distance to the edge, $\omega_b/(\Omega_{\min} - \omega)$. A third ratio, $\mu = \omega_b/\Delta\Omega$, compares trapping with the mixing of the whole band. It becomes the controlling parameter when the initial condition is far from any equilibrium (Section 3.4).

Dependence on the amplitude. In the linear regime $\delta\Phi \propto \varepsilon$. At $t = 0$ this is exact, because the perturbation of the weights and the Poisson equation are both linear in ε . Hence $\omega_b(\varepsilon) = \omega_b(1)\sqrt{\varepsilon}$, where $\omega_b(1)$ is the bounce frequency of the same perturbation at $\varepsilon = 1$, and

$$\nu(\varepsilon) = \nu_1 \sqrt{\varepsilon}, \quad \nu_1 = \frac{\omega_b(1)}{\gamma}.$$

The factor ν_1 depends on the equilibrium and on the shape of the perturbation, not on its amplitude. The ratio $\omega_b/(\Omega_{\min} - \omega)$ and μ scale in the same way, so their quotients are fixed by the equilibrium; for instance $\mu/\nu = \gamma/\Delta\Omega$. Two consequences follow. First, in a given equilibrium the amplitude changes all the nonlinear parameters together; changing one relative to the others requires another equilibrium, for instance another width J_t , which changes γ and $\Delta\Omega$ differently. Second, since $\varepsilon \leq 1$, $\nu \leq \nu_1(\eta)$. Far from the transition the coupling is weak, γ is large and ν_1 is small, so large values of ν may be out of reach; near η_c , $\gamma \rightarrow 0$ and any ν can be reached. On the discrete side the ratio $\omega_b/(\Omega_{\min} - \omega)$ is bounded by $\omega_b(1)/(\Omega_{\min} - \omega)$, which decreases away from the transition, so a mode far below the band cannot be made fragile with $\varepsilon \leq 1$.

The evolution is therefore controlled by three time scales: the mixing time τ_1 , the damping time $1/\gamma$, which is infinite for a discrete mode, and the bounce period $2\pi/\omega_b \propto \varepsilon^{-1/2}$.

1.8 Predictions

Three predictions follow from this framework and can be falsified by the runs of Section 3.3.

- P1. Where $\nu \ll 1$, the PIC evolution follows the linear theory of the same equilibrium for any ε .
- P2. Where $\nu \gtrsim 1$, the PIC evolution departs from linear theory at $t \sim 2\pi/\omega_b$, and damping stops.
- P3. A discrete mode becomes fragile when $\omega_b/(\Omega_{\min} - \omega) \gtrsim 1$.

P1 and P2 were formulated in the η battery, before the runs. P3 was formulated after the first two runs with discrete modes (D3 and D4) and tested blind with two later runs at larger mass (D9 and D10). For the non-equilibrium initial condition of Section 3.4 we expected a transition near $\mu \sim 1$.

2 Numerical methods

2.1 Self-consistent equilibria

The equilibrium of Section 1.3 is a fixed point: the action is computed in the total potential $\Phi = \Phi_{\text{iso}} + \Phi_{\text{self}}$, and Φ_{self} is the potential of $F_{\text{eq}}(J)$ itself. `equilibrio.py` finds it by iteration. The inputs are the shape of the distribution (g , W_0 and J_t), the angular momentum L_0 and the mass a_0 . The calculation uses its own radial grid, $r \in [0.01, 25]$ with spacing 0.01, ten times finer than the PIC grid.

1. Start from $\Phi_{\text{self}} = 0$, the isochrone alone.
2. Compute the angle-action map of the current potential. At fixed L the action depends only on the energy, so it suffices to tabulate $J(E)$. We take 4000 energies, from that of the circular orbit, $E_c = \Phi_{\text{ef}}(r_c)$, to one whose action slightly exceeds J_t . For each energy the turning points $r_{\pm}$, where $p_r = 0$, are found by bisection, and

$$J(E) = \frac{1}{\pi} \int_{r_-}^{r_+} p_r dr, \quad p_r = \sqrt{2(E - \Phi_{\text{ef}}(r))},$$

is computed with 64-point Gauss-Legendre quadrature. At a turning point $\Phi'_{\text{ef}}(r_{\pm}) \neq 0$, unless the orbit is circular, so $E - \Phi_{\text{ef}}$ vanishes linearly in r and $p_r \propto |r - r_{\pm}|^{1/2}$. The integrand of J is zero there, but its derivative is infinite. The map also needs the radial period, $2\pi/\Omega = 2 \int_{r_-}^{r_+} dr/p_r$, for the angle Q , and the integrand $1/p_r$ of the period is itself infinite at $r_{\pm}$. A Gauss-Legendre rule with n nodes integrates polynomials of degree up to $2n - 1$ exactly, so it converges fast only for smooth integrands, and slowly for these two. The substitution $r = r_m + r_a \sin \theta$, with $\theta \in [-\pi/2, \pi/2]$ and r_m , r_a the centre and half-width of $[r_-, r_+]$, makes both integrands smooth. Since $(r_+ - r)(r - r_-) = r_a^2 \cos^2 \theta$ and the zeros of $E - \Phi_{\text{ef}}$ at $r_{\pm}$ are simple, $E - \Phi_{\text{ef}} = r_a^2 \cos^2 \theta h(\theta)$ with h smooth and positive. Then $p_r = r_a \cos \theta \sqrt{2h}$, and the integrands of J and of the period become

$$p_r \frac{dr}{d\theta} = r_a^2 \cos^2 \theta \sqrt{2h}, \quad \frac{1}{p_r} \frac{dr}{d\theta} = \frac{1}{\sqrt{2h}}.$$

In the bare isochrone, where both integrals are known in closed form, 64 nodes give J to 10^{-14} and the period to 10^{-11} . Without the substitution the errors are 2×10^{-6} and 9×10^{-3} , and they decrease only as n^{-3} and n^{-1} . Inverting the table gives $E(J)$, and from it $\Omega(J) = dE/dJ$.

3. Evaluate the lowered Maxwellian in this potential. The cut-off energy is that of the edge orbit, $E_t = E(J_t)$, and the temperature is $T = (E_t - E_c)/W_0$. Both are recomputed at each iteration, so the support is always $[0, J_t]$ and the depth of the well is always W_0 temperatures.
4. Normalize. The map $(r, p_r) \rightarrow (Q, J)$ is canonical, $dr dp_r = dQ dJ$, and F_{eq} does not depend on Q , so

$$a_0 = 8\pi^2 L_0 \iint F_{\text{eq}} dQ dJ = 16\pi^3 L_0 A \int_0^{J_t} E_\gamma\left(g, \frac{E_t - E(J)}{T}\right) dJ,$$

which fixes A . The mass is exactly a_0 at every iteration; what changes from one iteration to the next is the shape of F_{eq} in J .

5. Compute the density by integrating over the allowed radial velocities at each radius,

$$4\pi r^2 \rho(r) = 8\pi^2 L_0 \int F_{\text{eq}}(E(r, p_r)) dp_r, \quad E = \frac{1}{2}p_r^2 + \Phi_{\text{ef}}(r).$$

Since F_{eq} is a function of the energy, this step does not need the map.

6. Solve the Poisson equation in spherical symmetry, $\Phi_{\text{self}}(r) = -M(< r)/r - \int_r^\infty 4\pi r' \rho(r') dr'$.
7. Repeat steps 2 to 6 until $\max_r |\Phi_{\text{self}}^{(n+1)} - \Phi_{\text{self}}^{(n)}| < 10^{-13}$. No relaxation is used.

The result, $\Phi_{\text{self}}(r)$, $\rho(r)$, the table $E(J)$ and the amplitude A , is stored in `exe/demo_eta/ic/<run>_equilibrio.npz`. The table gives the frequency band and the angle-action map used by all diagnostics.

Convergence of the iteration. Steps 2–6 define a map $\mathcal{T}$ on the potential, $\Phi_{\text{self}}^{(n+1)} = \mathcal{T}(\Phi_{\text{self}}^{(n)})$, and the equilibrium is its fixed point. By the Banach fixed-point theorem, if $\mathcal{T}$ is a contraction near the solution, $\|\mathcal{T}(\Phi) - \mathcal{T}(\Phi')\| \leq q \|\Phi - \Phi'\|$ with $q < 1$, the iteration converges geometrically from any starting point in that neighbourhood, the fixed point is unique there, and the error of the last iterate is bounded by $q/(1 - q) \|\Phi_{\text{self}}^{(n)} - \Phi_{\text{self}}^{(n-1)}\|$. For small mass the contraction holds: Φ_{self} is the potential of a fixed mass a_0 , and a change of Φ only changes how that mass is distributed in r , so $\mathcal{T}$ depends on Φ at order a_0 and $q = O(a_0)$. The same argument, through the implicit function theorem around the isochrone alone, gives existence and uniqueness of the equilibrium for small a_0 . For finite mass there is no a priori proof, so q is measured: `equilibrio.py` takes the median ratio of consecutive changes while both exceed 10^{-10} , far above round-off, and reports the bound. Table 1 gives the values. In each case the ratio is constant to three digits during the iteration (from 0.174 to 0.176 in A4), the signature of a clean contraction, and for the six equilibria with $g = 2$ it follows $q \approx a_0/(a_0 + 0.35)$ to within 0.002. The bound on the error of the final potential is below 10^{-13} in all cases. The script stops with an error if the change is still above the tolerance after 60 iterations. By the fit this would happen near $a_0 \approx 0.7$, where $q \approx 0.67$ and about 68 iterations are needed; larger masses would require more iterations or an accelerated iteration (Anderson mixing).

An independent construction. For the lowered Maxwellian, F depends on the energy only, so the density at radius r depends only on the potential at r . With $\psi(r) = (E_t - \Phi_{\text{ef}}(r))/T$, integrating over $|p_r| \leq \sqrt{2T\psi}$ gives

$$4\pi r^2 \rho(r) = C E_\gamma\left(g + \frac{1}{2}, \psi(r)\right), \quad C = 8\pi^2 L_0 A \sqrt{2\pi T},$$

case	a_0	iterations	q	bound on the final error
A1	0.0065	7	0.017	1.0×10^{-16}
A3	0.042	12	0.105	2.6×10^{-15}
A4	0.075	15	0.175	1.1×10^{-14}
L5	0.097	17	0.216	1.7×10^{-14}
L6	0.170	24	0.327	1.6×10^{-14}
G1a ($g = 1$)	0.0069	7	0.017	9.6×10^{-17}
M5	0.5	50	0.587	9.6×10^{-14}

Table 1: Convergence of the fixed-point iteration for the seven equilibria. q is the contraction factor and the last column is $q/(1-q)$ times the last change. The bound refers to the fixed point of the discretized map; the discretization error itself is measured below.

and $\rho = 0$ where $\psi \leq 0$, because $\int E_\gamma(g, \psi - p^2/2T) dp = \sqrt{2\pi T} E_\gamma(g + \frac{1}{2}, \psi)$, term by term in the series of E_γ with the Beta function $B(g + n + 1, \frac{1}{2})$. Poisson's equation then becomes an ordinary differential equation. With $u = \Phi_{\text{self}} - \Phi_{\text{self}}(0)$ and $\tilde{E} = E_t - \Phi_{\text{self}}(0)$,

$$\frac{dM}{dr} = C E_\gamma(g + \frac{1}{2}, \psi), \quad \frac{du}{dr} = \frac{M}{r^2}, \quad \psi = \frac{\tilde{E} - \Phi_{\text{iso}}(r) - L_0^2/2r^2 - u}{T}.$$

The centrifugal barrier leaves the centre empty, so $M = u = 0$ up to the inner edge of the support, and outside the support $\Phi_{\text{self}} = -M/r$, which fixes $\Phi_{\text{self}}(0)$. The system is integrated only over the support $[r_-, r_+]$, where $\psi > 0$ and the right-hand side is smooth, with the eighth-order Dormand–Prince method (relative tolerance 10^{-12}). The three numbers $(\tilde{E}, T, C)$ are fixed by three conditions: the total mass, $M(r_+) = a_0$; the depth, $\psi(r_c) = W_0$ at the circular orbit (the minimum of Φ_{ef}), which is $T = (E_t - E_c)/W_0$; and the edge, $J(E_t) = \frac{1}{\pi} \int_{r_-}^{r_+} \sqrt{2T\psi} dr = J_t$, since the turning points of the edge orbit are the edges of the support. They are found with a Powell hybrid root finder (`scipy.optimize.root`), starting from the isochrone alone, with continuation in the mass if it fails. Each equilibrium takes about 0.5 s. This construction has no iteration on a field: its error is set by the integrator tolerance, and existence and uniqueness reduce to a root problem in three variables. It applies only to distributions that depend on E , such as the Wilson and King models; the Gaussian and the hollow family of the η battery depend on J , which is not local in the potential. It is implemented in `equilibrio_edo.py` and selected with `equilibrio.py --metodo edo`. In the limit $a_0 = 10^{-9}$ it reproduces the isochrone values of E_t and E_c to 1.4 and 1.6×10^{-10} , the self-potential of that mass ($\approx -a_0/r_c$), and T to 7×10^{-12} .

Accuracy of the two constructions. Table 2 compares the two for the seven equilibria (`comparar_equilibrios.py`). They agree to $1\text{--}2 \times 10^{-6}$ in Φ_{self} , relative to its maximum, and to 10^{-5} or better in the density and the parameters. The solution of the differential equation is the reference: it changes by 4×10^{-10} when its tolerance is relaxed a hundredfold and its quadrature nodes are halved. The difference is the discretization error of the iteration. It is dominated by the radial grid: with twice as many energies and momenta it is unchanged (1.1×10^{-6} in A4), and with radial spacings $h = 0.01, 0.005$ and 0.0025 it decreases as h^2 , to 1.1×10^{-6} , 2.8×10^{-7} and 6.5×10^{-8} , at the same $q = 0.175$. The mass of the iteration, integrated on its grid, differs from a_0 by at most 1.9×10^{-7} (M5). Both constructions converge to the same solution, and the remaining error, $\sim 10^{-6}$, is far below that of the PIC discretization (Section 5.2).

Stationarity in the PIC code. The final test of an equilibrium is to evolve it with the code at $\varepsilon = 0$. The equilibria built by the two methods evolve in the same way, to better than 1 %, and the departures from stationarity are set by the PIC discretization, not by the construction (Section 5.2).

case	Φ_{self}	ρ	E_t	T	A
A1	9.3×10^{-7}	1.2×10^{-7}	3.3×10^{-9}	8.1×10^{-8}	6.2×10^{-8}
A3	1.0×10^{-6}	7.9×10^{-7}	2.1×10^{-8}	5.5×10^{-7}	4.2×10^{-7}
A4	1.1×10^{-6}	1.5×10^{-6}	4.0×10^{-8}	1.0×10^{-6}	7.7×10^{-7}
L5	1.2×10^{-6}	1.9×10^{-6}	5.2×10^{-8}	1.3×10^{-6}	1.0×10^{-6}
L6	1.4×10^{-6}	3.5×10^{-6}	9.7×10^{-8}	2.4×10^{-6}	1.8×10^{-6}
G1a	9.5×10^{-7}	1.8×10^{-7}	4.8×10^{-9}	7.7×10^{-8}	4.3×10^{-8}
M5	2.3×10^{-6}	1.2×10^{-5}	3.6×10^{-7}	8.2×10^{-6}	6.0×10^{-6}

Table 2: Relative differences between the equilibrium of the fixed-point iteration and that of the differential equation: maximum over the grid of $|\Delta\Phi_{\text{self}}|$ relative to $\max|\Phi_{\text{self}}|$, the same for ρ , and the relative differences of E_t , T and A .

2.2 Particle loading

The particles sit at the nodes of a regular grid in the angle-action variables of the converged equilibrium, $J_i = (i - \frac{1}{2})J_t/N_{rc}$ and $Q_j = (j - \frac{1}{2})2\pi/N_{pc}$, with $N_{rc} = 400$ and $N_{pc} = 25$, so $N = 10^4$. For each node the map is inverted. With $E = E(J_i)$ and its turning points, bisection finds the point of the orbit at which the fraction of the radial period elapsed since pericentre is $Q_j/2\pi$. This gives r , and $p_r = \pm\sqrt{2(E - \Phi_{\text{ef}}(r))}$, with the positive sign for $Q_j \leq \pi$. The forward map applied to the resulting (r, p_r) returns the nodes with errors below 10^{-8} in J and 5×10^{-7} in Q . The weight of each particle is $F_{\text{eq}}(J_i) [1 + \varepsilon s(J_i) \cos Q_j]$. When it reads the file, the code multiplies all weights by a common constant so that the total mass is a_0 ; since all grid cells have the same phase-space volume, this only fixes the normalization. The perturbation changes neither the mass nor the mean profile in J , because $\sum_j \cos Q_j = 0$ exactly on a regular grid. This is a quiet start: the perturbation is carried by the weights, not by the positions, and there is no sampling noise. The reference runs (Z) use the same nodes with $\varepsilon = 0$.

The grid must be built in the self-consistent potential. The code's own `aa_quad` state builds it in the variables of the bare isochrone, and then, once the mass is no longer negligible, the initial condition is not stationary: the particles are not where the actual potential would keep them. With the present procedure the action of each particle in the unperturbed runs changes by at most $1.2 \times 10^{-4}J_t$ (A1) to $9.5 \times 10^{-3}J_t$ (M5, the most massive case), less than 1 % of J_t (Section 5.2).

2.3 PIC scheme

The PIC code (VP_PIC, Fortran with OpenMP) represents the distribution by N particles. Each particle is a spherical shell with a fixed mass (its weight), a radius r_i and a radial momentum p_i . The radial grid is uniform and staggered, $r_j = (j - \frac{1}{2})\Delta r$ for $j = 1, \dots, N_r$, with $\Delta r = 0.1$ and $r \in [0, 20]$ ($N_r = 200$). The self-gravity is computed in four steps.

1. Deposition. The mass of each particle is assigned to the neighbouring grid points with linear B-spline (cloud-in-cell) weights and divided by the cell volume.
2. Density. Between grid points the density is taken to be linear, with values rescaled so that the field sees exactly the deposited mass.
3. Field. The enclosed mass and the potential follow from $dM/dr = 4\pi r^2 \rho$ and $d\Phi_{\text{self}}/dr = M/r^2$, integrated in closed form over each cell. The density is even at the origin. The additive constant is fixed by $\Phi_{\text{self}} + r d\Phi_{\text{self}}/dr = 0$ at the outer grid point, which matches the exterior solution $-M/r$.
4. Interpolation. The potential and the force are interpolated back to the particles with the same weights. Beyond the last grid point the exterior solution, $\Phi \propto 1/r$ and $F \propto 1/r^2$, is

used.

The isochrone force and the centrifugal force L_0^2/r^3 are evaluated analytically at each particle. The equations of motion $\dot{r} = p_r$, $\dot{p}_r = -\partial_r H$ are integrated with the fourth-order symplectic scheme of Yoshida (1990): a symmetric composition of three kick–drift–kick leapfrog sub-steps with weights w_1 , w_0 , w_1 , where $w_1 = 1/(2 - 2^{1/3})$ and $w_0 = -2^{1/3}/(2 - 2^{1/3})$. Each sub-step includes a Poisson solve. The time step is $\Delta t = C \Delta r/p_{\max} = 0.05$, with Courant factor $C = 1$ and $p_{\max} = 2$. A particle that crosses the origin is reflected ($r \rightarrow -r$, $p_r \rightarrow -p_r$), but with $L_0 = 2$ the centrifugal barrier keeps the particles away from it. Positions, momenta, weights, the grid potential and the total energy (kinetic, background, centrifugal and self-gravitational) are written to HDF5 every 200 steps, that is, every 10 time units.

2.4 Linearized solver

The linear reference is computed by `lineal.py`, which integrates the linearized equation of Section 1.5 about the same equilibrium, without particles. It uses a fixed grid of $N_J \times N_Q = 1600 \times 32$ nodes in (Q, J) ; the radius $r(Q, J)$ of each node is computed once with the inverse equilibrium map. Each time step, of size $\Delta t = 0.5$, is a Strang splitting: half a field kick, $\delta F \leftarrow \delta F + \frac{1}{2} \Delta t F'_{\text{eq}}(J) \partial_Q \delta \Phi$, exact free transport in Q , which in Fourier space is $\hat{f}_k \rightarrow \hat{f}_k e^{-ik\Omega(J)\Delta t}$, and a second half kick. The derivative ∂_Q is computed by FFT. The potential is obtained with the Green’s function of spherical shells, which is exact at the nodes,

$$\delta \Phi(r_i) = -\frac{M(< r_i)}{r_i} - \sum_{r_j > r_i} \frac{m_j}{r_j}, \quad m_j = 8\pi^2 L_0 \delta F_j \Delta Q \Delta J,$$

with the shell of node i counted with half its mass. The initial condition is the perturbation of the PIC runs with $\varepsilon = 1$. The solver returns $\delta \Phi_{\text{lin}}(r, t)$ and $h_{1,\text{lin}}(t)$ up to the final time of the longest run on each equilibrium.

2.5 Diagnostics

Potential perturbation. At each snapshot we compute, on the code grid,

$$\delta \Phi(r, t) = \Phi(r, t) - \Phi_{\text{iso}}(r) - \Phi_{\text{self,eq}}(r).$$

This quantity does not depend on any angle–action map, and it is the one whose decay the theorems address. The part due to the perturbation is isolated by subtracting the reference run of the same equilibrium and dividing by ε ,

$$\delta \Phi_\varepsilon(r, t) = \frac{\delta \Phi(r, t; \varepsilon) - \delta \Phi(r, t; 0)}{\varepsilon},$$

which is independent of ε in the linear regime. Its size $\|\delta \Phi\|$ is the rms value over $3 \leq r \leq 15$, where the component lives.

Harmonic. With (Q_j, J_j) from the equilibrium map,

$$h_1(t) = \frac{\sum_j f_j B(J_j) e^{-iQ_j}}{\sum_j f_j B(J_j)|_{t=0}}, \quad B(J) = J^2 e^{-(J-0.05)^2/0.05^2},$$

with the same subtraction of the reference run and division by ε .

Linear reference and nonlinear deviation. Each PIC snapshot is paired with the nearest sample of the linear solution $\delta\Phi_{\text{lin}}$, $h_{1,\text{lin}}$ (Section 2.4). The nonlinear deviation is

$$\mathcal{D}(t) = \frac{\|\delta\Phi_\varepsilon - \delta\Phi_{\text{lin}}\|}{\|\delta\Phi_{\text{lin}}\|}.$$

In practice we use envelopes (below): t_{nl} is the first time at which the envelope of $\|\delta\Phi_\varepsilon - \delta\Phi_{\text{lin}}\|$ exceeds 10 % of the envelope of $\|\delta\Phi_{\text{lin}}\|$.

Envelopes, ratios and losses. The envelope of a signal is its maximum over windows of ± 50 time units, slightly more than one radial period; without it the oscillation of the mode hides the trend. R is the ratio of the PIC and linear envelopes of $\|\delta\Phi\|$ in a time window; its error is half the difference between the values in the two halves of the window. The loss of a run between 5 and $20\tau_1$ is the relative decrease of the PIC envelope over that interval, net of the decrease of the linear envelope: $1 - [e_{\text{PIC}}(20\tau_1)/e_{\text{PIC}}(5\tau_1)]/[e_{\text{lin}}(20\tau_1)/e_{\text{lin}}(5\tau_1)]$.

Pole fits. With e^{-iQ_j} in its definition, h_1 picks the $k = 1$ part of the mode, so $h_1 \propto e^{-i(\omega - i\gamma)t}$ with $\omega > 0$. The pole $\omega - i\gamma$ is fitted, in the PIC runs and in the linear solution alike, with the matrix-pencil method of order $M = 3$ in windows of a few τ_1 . Its uncertainty is the largest deviation among the fits with $M = 2$ and $M = 3$ on three trims of the window (complete, without the first 10 %, and without the last 10 %). This bounds the error of the fit, not the numerical error of the run, which is assessed with the convergence tests (Section 5). We tested the fit with synthetic signals of known γ at $\omega = 0.0725$, sampled every 10 time units and fitted over $3600 \leq t \leq 7200$ ($10\text{--}20\tau_1$ with $\tau_1 = 360$): the frequency, sampling and window of the discrete mode of L5. Both for clean signals and with a 5 % beat and complex noise of amplitude 10^{-3} , the fit recovers γ to $\sim 10^{-6}$ for $0 \leq \gamma \leq 10^{-4}$ (`demo_eta.py` `resolucion`).

Smooth and fine parts. To separate the large-scale response from structure on the scale of a few grid cells, $\delta\Phi_\varepsilon(r)$ is smoothed with a Gaussian of width 0.5 in r . The smooth part is compared with the linear solution smoothed in the same way, and the remainder, relative to the initial linear norm, is the fine part. A scale-free alternative is the projection of $\delta\Phi_\varepsilon(r)$ on the profile of the linear solution at $t = 5\tau_1$. The noise floor of a run is estimated as the rms fluctuation of $\delta\Phi$ of its reference run about its time mean, divided by ε .

Trapped particles. A particle is counted as trapped in a time window if its phase relative to the wave, $\psi = Q - \omega t$, librates, that is, varies by less than 2π , instead of circulating. The amplitude of the wave felt at the resonance is $A(t) = |\delta\Phi_1(J_r, t)|$, the first Fourier coefficient in Q of $\delta\Phi$ along the equilibrium orbit of action J_r .

Phase-space figures. Each particle is plotted in (r, p_r) and in (Q, J) , coloured by its weight perturbation $f_j - f_{\text{eq},j} = \varepsilon F_{\text{eq}} s \cos Q_0$: red where the perturbation adds mass and blue where it removes it. The weights do not change, so the colour is a label carried by the particle, and its pattern is the fine structure of δF . In the Z runs the colour is $\cos Q_0$, a label without dynamical effect. In (Q, J) each row of fixed action advances with its $\Omega(J)$, so the initial pattern of columns tilts, and phase mixing appears directly as shear.

3 Linear predictions and run design

3.1 Linear theory of the equilibrium family

All equilibria are Wilson models with $J_t = 0.138$, $W_0 = 3$, $L_0 = 2$ and $g = 2$, except one King model ($g = 1$, G1a) used to test the vacuum edge. In this family η grows slightly more slowly

than linearly with the mass: η/a_0 decreases from 15 to 12 between $a_0 = 0.0065$ and 0.17, and is 10 at $a_0 = 0.5$. The value $\eta = 1$ corresponds to $a_0 = 0.075$.

The linearized solver (Section 2.4) gives the pole $\omega - i\gamma$ of each equilibrium (Table 3 and Figure 1). As η increases the mode moves down towards $\Omega_{\min}$ and is damped less and less. Between $\eta = 1$ and 1.24 it crosses the edge and becomes a discrete mode, so $\eta_c \approx 1.1$ –1.2 for this family. The full linear analysis, including the scan in η and its checks, is in the companion document on the η battery (docs/experimento_eta/bateria_eta.pdf).

The solver also reproduces the edge dichotomy of Section 1.6. At $\eta = 1$, $g \leq 1$ already has a discrete mode while $g \geq 2$ is still damped. At $\eta = 0.1$ the sharp edge (G1a) leaves a response that oscillates at $\Omega_{\min}$ and decays ever more slowly, while the response of the smooth edge (A1) dies out.

In A4, for example, the band is $[0.0698, 0.0857]$, so $\Delta\Omega = 0.0159$ and $\tau_1 = 395$. The radial period ranges from $2\pi/0.0857 = 73$ at the circular orbit to $2\pi/0.0698 = 90$ at the edge of the support, and the two ends of the band dephase by one full turn after about five radial periods. Across the family τ_1 ranges from 125 (M5) to about 560 (A1), and the damping time $1/\gamma$ is 170 in A1, 420 in A3 and 2700 in A4.

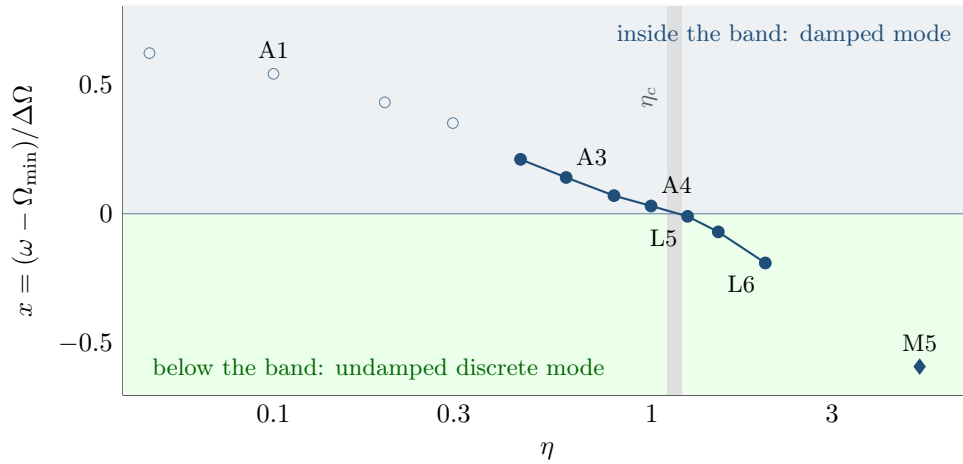


Figure 1: Position of the linear mode relative to the band as a function of η , for the Wilson family with $g = 2$. As η increases the mode moves down to the lower edge of the band ($x = 0$) and, beyond $\eta_c \approx 1.1$ –1.2 (grey strip), leaves it as a discrete mode. Open symbols ($\eta \leq 0.3$): there is no isolated pole and x is approximate. Labels mark the equilibria used in the runs.

case	a_0	η	band $[\Omega_{\min}, \Omega_{\max}]$	x	$\gamma/\Delta\Omega$	linear regime
A1	0.0065	0.10	$[0.0610, 0.0723]$	~ 0.5	~ 0.5	fast damping
A3	0.042	0.59	$[0.0655, 0.0791]$	0.14	0.17	damped
A4	0.075	1.00	$[0.0698, 0.0857]$	0.03	0.023	damped, at the edge
L5	0.097	1.24	$[0.0727, 0.0902]$	-0.01	0	discrete mode next to the edge
L6	0.170	2.00	$[0.0828, 0.1055]$	-0.19	0	discrete mode, separated
M5	0.5	5.11	$[0.1342, 0.1843]$	-0.59	0	mass comparable to the background; mode far from the band
G1a	0.0069	0.10	$[0.0610, 0.0723]$	≈ 0	–	sharp edge ($g = 1$): persistent response

Table 3: The equilibria and their linear pole (from the scan of the η battery). All are Wilson models with $J_t = 0.138$, $W_0 = 3$, $L_0 = 2$ and $g = 2$, except G1a, a King model ($g = 1$).

3.2 Choice of η and ν

The two parameters depend on different inputs. η depends only on the equilibrium, that is, on the mass and the shape of the distribution. ν depends on the equilibrium and on the amplitude of the perturbation. They can therefore be set independently: first the equilibrium is chosen for its η , then the amplitude for the desired ν .

η is set by the mass. With the converged equilibrium, `eta.py` computes $\Omega(J) = dE/dJ$ at 4000 points in $[0, J_t]$ and the mass-weighted mean frequency

$$\bar{\Omega} = \frac{\int_0^{J_t} F_{\text{eq}} \Omega dJ}{\int_0^{J_t} F_{\text{eq}} dJ}.$$

It repeats the calculation with $a_0 = 0$, the same shape in the isochrone alone, to obtain $\bar{\Omega}_{\text{iso}}$, and forms

$$\eta = \frac{(\bar{\Omega} - \bar{\Omega}_{\text{iso}})/\bar{\Omega}_{\text{iso}}}{(\Omega_{\text{max}} - \Omega_{\text{min}})/\bar{\Omega}}.$$

The amplitude ε does not enter. There is no closed inverse, but η increases monotonically with a_0 (η/a_0 decreases from 16 to 10 between $a_0 = 0.003$ and 0.5), so a given η is obtained by scanning a_0 and interpolating: $a_0 = 0.075$ gives $\eta = 0.996$ (A4). Changing J_t , W_0 or g at fixed mass also changes η .

ν is set by the amplitude. `lineal.py` gives the pole $\omega - i\gamma$ of the equilibrium. The bounce frequency is estimated from the potential of the perturbation at $t = 0$, which the linear solution with $\varepsilon = 1$ provides:

$$\omega_b(\varepsilon) = \sqrt{|\Omega'| \max_r |\delta\Phi(r, 0; \varepsilon)|}, \quad |\Omega'| \approx \frac{\Delta\Omega}{J_t}.$$

Since $\nu = \nu_1 \sqrt{\varepsilon}$ (Section 1.7), the amplitude that gives a chosen ν is $\varepsilon = (\nu/\nu_1)^2$. The runs were set in this way. For D2, in A3, $\nu_1 = 1.17$, and $\nu = 0.3$ requires $\varepsilon = (0.3/1.17)^2 = 0.066$ (0.065 was used). For D5 and D6, in A4, $\nu_1 = 10.9$, and $\nu = 3$ and $\nu = 10$ require $\varepsilon = 0.075$ and 0.84. For a discrete mode the ratio $\omega_b/(\Omega_{\text{min}} - \omega)$ plays the role of ν . The amplitudes of the ε scan of D3 were chosen with it: in L5 the factor is 19, and $\varepsilon = 0.003, 0.01, 0.03$ and 0.1 give 1.0, 1.9, 3.3 and 6.0. For μ the factor is $\mu_1 = \omega_b(1)/\Delta\Omega$, 0.25 in A4. Table 4 lists these factors for the whole family.

Varying one parameter at a time. To vary ν at fixed η , the same equilibrium is run at several amplitudes, as D5 and D6 in A4 or the ε scan of D3. To vary η at fixed ν , the mass must be changed *and* the amplitude readjusted in each equilibrium, $\varepsilon = (\nu/\nu_1(\eta))^2$, because ν_1 grows by a factor of about sixty between $\eta = 0.1$ and 1. A scan in η at fixed ε also changes ν : at $\varepsilon = 0.075$, ν goes from 0.05 at $\eta = 0.1$ to 3 at $\eta = 1$. Keeping $\nu = 1$ requires $\varepsilon = 0.73$ at $\eta = 0.6$, 0.098 at $\eta = 0.8$ and 0.0084 at $\eta = 1$.

Since $\mu/\nu = \gamma/\Delta\Omega$ is fixed by the equilibrium (Section 1.7), in A4, where $\gamma/\Delta\Omega = 0.023$, $\mu = 0.023\nu$ at any amplitude: D5 has $\nu = 3$ and $\mu = 0.07$, and D6 has $\nu = 10$ and $\mu = 0.23$.

The ceiling in practice. Since $\varepsilon \leq 1$, and `equilibrio.py` refuses to build data with $F < 0$, part of the (η, ν) plane is inaccessible (Figure 2): for $\eta \leq 0.45$ this perturbation cannot reach $\nu = 1$. On the discrete side the largest ratio, $\omega_b(1)/(\Omega_{\text{min}} - \omega)$, is 19 in L5, 1.7 in L6 and 0.63 in M5, so the mode of M5 cannot be made fragile with $\varepsilon \leq 1$, which is what D10 shows. The ceiling can be raised at fixed η in three ways: with a perturbation that produces more $\delta\Phi$ per unit ε , through another radial profile $s(J)$ or angular profile $\chi(Q)$; with a different width J_t at

equilibrium	a_0	η	linear scale Λ	$\omega_b(\varepsilon = 1)$	$\omega_b(1)/\Lambda$	ε for ratio 1
<i>mode inside the band: $\Lambda = \gamma$, and the ratio is ν_1</i>						
L1	0.0030	0.05	7.4×10^{-3}	6.7×10^{-4}	0.09	—
A1	0.0065	0.10	6.0×10^{-3}	1.0×10^{-3}	0.17	—
L2	0.013	0.20	4.7×10^{-3}	1.4×10^{-3}	0.31	—
A2	0.020	0.30	4.0×10^{-3}	1.8×10^{-3}	0.46	—
L3	0.031	0.45	3.3×10^{-3}	2.3×10^{-3}	0.71	—
A3	0.042	0.59	2.4×10^{-3}	2.8×10^{-3}	1.2	0.73
L4	0.058	0.80	1.1×10^{-3}	3.4×10^{-3}	3.2	0.098
A4	0.075	1.00	3.7×10^{-4}	4.0×10^{-3}	10.9	0.0084
<i>discrete mode: $\Lambda = \Omega_{\min} - \omega$, the distance from the mode to the edge of the band</i>						
L5	0.097	1.24	2.5×10^{-4}	4.8×10^{-3}	19	0.0028
A5	0.121	1.50	1.4×10^{-3}	5.6×10^{-3}	4.1	0.060
L6	0.170	2.00	4.3×10^{-3}	7.2×10^{-3}	1.7	0.36
M5	0.5	5.11	2.9×10^{-2}	1.8×10^{-2}	0.63	—

Table 4: Factor that converts the amplitude into the nonlinear parameter, for the Wilson family with $J_t = 0.138$, $W_0 = 3$, $g = 2$ and $L_0 = 2$. At amplitude ε the ratio is $(\omega_b(1)/\Lambda)\sqrt{\varepsilon}$. The last column is the amplitude that makes it equal to 1; “—” means that $\varepsilon > 1$ would be needed, which this perturbation does not allow. For $\eta \leq 0.3$ there is no isolated pole and γ is only an order of magnitude. Data from `barrido_lineal.py`.

fixed η , which changes γ , $\Delta\Omega$ and ν_1 , although η_c moves as well; or with a non-perturbative initial condition, such as the blob of Section 3.4, whose amplitude is of order one by construction. The blob, however, is not a perturbation of any equilibrium, and the parameter that orders its evolution is μ .

Accuracy of the design value of ν . It is an estimate used to order the runs and choose ε , not a precise threshold. It uses the maximum of $|\delta\Phi|$ over all radii and the mean slope of $\Omega(J)$. A resonant particle feels instead the Fourier coefficient of $\delta\Phi$ along its orbit, which is smaller, and the amplitude decays with time. In D6 the design bounce period was 1700; with the amplitude measured at the resonance the pendulum gives 7700, and the amplitude oscillates with a period of ~ 9000 (Section 4.2). The design value of ν overestimates the effective one by a factor of a few.

3.3 Runs

Ten runs cover the regimes of Section 1: both sides of the transition at low amplitude, the edge of the transition at increasing amplitude, the discrete mode at large amplitude, the sharp edge, and a mass comparable to that of the background (Table 5 and Figure 2). Each equilibrium also has a reference run with $\varepsilon = 0$ (the Z runs), which is subtracted to isolate the response to the perturbation. There are seventeen simulations in total.

The damped runs in the linear regime last $10\text{--}20\tau_1$, enough to see the tail after mixing. For D5 and D6 the relevant scale is the bounce period $2\pi/\omega_b$, 5700 and 1700: both runs cover about three bounce periods, the time over which O’Neil’s theory predicts that damping stops and the amplitude oscillates. These periods were estimated with the initial amplitude. Since the amplitude decays, the actual period of D6 turned out to be ~ 9000 , and the run D6L, extended to $t = 17\,200$, was added to measure it (Section 4.2).

All runs use the PIC scheme of Section 2.3 with $N = 400 \times 25 = 10^4$ particles, and were run one after another with four OpenMP threads. The `.par` files are in `reproducir/corridas/12_demo_eta/` and the whole workflow in `reproducir/scripts/demo_eta.py` (Appendix E). The control runs of Section 5 differ from these only in N , Δt , ε or the final time.

run	equil.	η	g	ε	ν	μ	t_{fin}	t_{fin}/τ_1	regime tested
D1	A1	0.10	2	1	0.17	0.09	11100	20	weak coupling: linear even at $\varepsilon = 1$
D2	A3	0.59	2	0.065	0.30	0.05	4600	10	damped, linear
D3	L5	1.24	2	0.1	∞	0.09	7200	20	discrete mode next to the edge
D4	L6	2.00	2	0.1	∞	0.10	5520	20	discrete mode separated from the band
D5	A4	1.00	2	0.075	3.0	0.07	17200	44	edge of the transition, nonlinear
D6	A4	1.00	2	0.84	10	0.23	5200	13	edge of the transition, strongly nonlinear
D7	L5	1.24	2	1	∞	0.27	7200	20	the mode of D3 at large amplitude
D8	G1a	0.10	1	1	~ 2	0.11	11140	20	sharp edge (King)
D9	M5	5.11	2	0.1	∞	0.12	5000	40	mass comparable to the background
D10	M5	5.11	2	1	∞	0.37	5000	40	the same, large amplitude

Reference runs with $\varepsilon = 0$: Z_A1 (for D1), Z_A3 (D2), Z_A4 (D5 and D6), Z_L5 (D3 and D7), Z_L6 (D4), Z_G1a (D8) and Z_M5 (D9 and D10).

Table 5: The seventeen runs. $\nu = \omega_b/\gamma$ and $\mu = \omega_b/\Delta\Omega$, with $\omega_b \propto \sqrt{\varepsilon}$ estimated from the linear solution at $t = 0$; $\nu = \infty$ denotes a discrete mode ($\gamma = 0$). In G1a the late decay is not exponential and γ is ill-defined, so its ν is only indicative. Each reference run lasts as long as the longest run it accompanies.

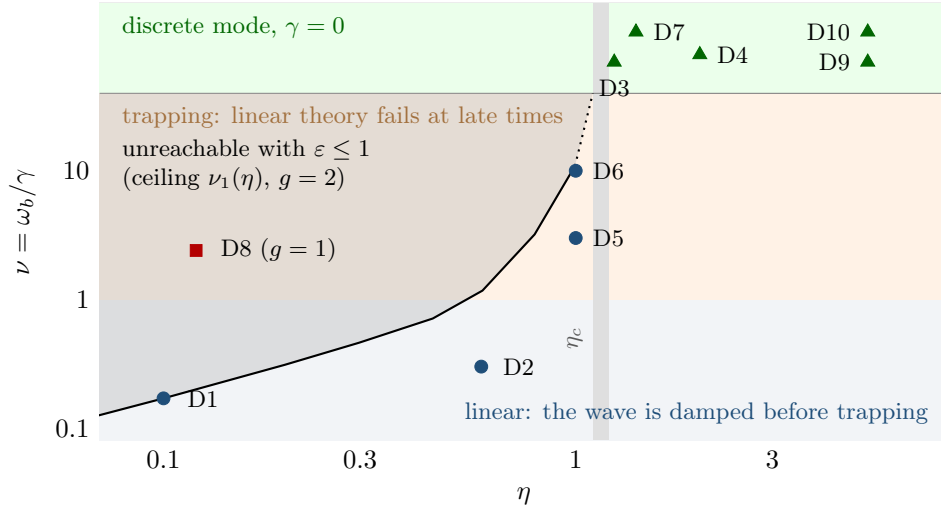


Figure 2: The runs in the (η, ν) plane. The grey strip is the linear transition, $\eta_c \approx 1.1$ – 1.2 . The black curve is the ceiling $\nu_1(\eta) = \omega_b(\varepsilon = 1)/\gamma$ of the $g = 2$ family (Table 4; dotted, the extrapolation towards η_c , where $\gamma \rightarrow 0$). With this perturbation no run can lie above it, and the shaded region is unreachable (Section 3.2). D1 lies on the ceiling ($\varepsilon = 1$). D8 belongs to another family ($g = 1$), with a higher ceiling of its own because its late decay is slow, and is shifted slightly in η from D1 for legibility. The discrete modes are placed at the top, off scale: there are no resonant particles and ν does not apply. D3 and D7 have the same η and are separated for legibility, as are D9 and D10. The η axis is logarithmic.

3.4 Non-equilibrium initial condition: the blob

The runs of Section 3.3 start from an equilibrium plus a perturbation. The initial condition of the earlier self-gravity runs (`reproducir/corridas/09_autogravedad`) is different: a blob in angle,

$$F_0(Q, J) = e^{-\sin^2(Q/2)/s_Q^2} J^2 e^{-J^2/s_J^2}, \quad s_Q = s_J = 0.1,$$

generated by the code itself (`state = aa_quad`, `dftype = gauss`) on a regular grid in the angle-action variables of the *bare* isochrone. All the mass is concentrated in one orbital phase, with a width of 0.2 rad. In the terms of Section 1.3 it is a perturbation with $\varepsilon \gg 1$ in all harmonics up to $k \sim 1/s_Q \approx 10$, and there is no equilibrium of which it is a perturbation. The earlier runs used $a_0 = 10^{-4}$ – 10^{-2} .

Without self-gravity the blob is sheared, and its harmonics h_k tend to zero by phase mixing. With self-gravity its own field could hold it together. The scale that competes with the shear is the bounce frequency of the particles in the well of the blob itself, and the relevant ratio is $\mu = \omega_b/\Delta\Omega$ (Section 1.7): for $\mu \ll 1$ the particles dephase before they feel the well, and for $\mu \gtrsim 1$ they are trapped in it. In runs D1–D10, μ never exceeds 0.37.

We ran the same initial condition, on the same grid as the earlier runs ($N_{rc} \times N_{pc} = 400 \times 25$, $J \in [0, 0.6]$), at five masses, plus one run without self-gravity (free mixing) and one with four times more resolution in angle (Table 8). The runs use $\Delta t = 0.05$ and $t_{\text{fin}} = 10\,000 \approx 31 \tau_1$, where $\tau_1 = 327$ is that of the band of the bare isochrone for this distribution, $[0.0513, 0.0705]$. ω_b is estimated from the oscillation of the self-potential during the first radial period, $\omega_b = \sqrt{|\Omega'| |\delta\Phi|}$ with $|\Omega'| \approx \Delta\Omega/\Delta J$. Since the initial condition is not stationary, there is no fixed equilibrium to serve as a frame. The harmonics $h_k = \sum_j f_j e^{-ikQ_j} / \sum_j f_j$ are computed with two maps: that of the instantaneous potential, which describes the actual orbits, and that of the bare isochrone, which the code and the earlier runs use. The two maps agree at large mass but not at small mass (Section 4.5).

4 Results

The numbers of this section are in the files `resumen.txt` and `controles.txt` of `exe/demo_eta/`. The pole uncertainties are in Table 6.

4.1 Low amplitude: comparison with linear theory

run	window	PIC		linear		R	t_{nl}/τ_1
		ω	$\gamma [10^{-5}]$	ω	$\gamma [10^{-5}]$		
D1 ($\eta = 0.1$)	1–3 τ_1	0.0657(20)	259 ± 40	0.0657(20)	257 ± 30	0.999(10)	3.0
D2 ($\eta = 0.6$)	1–3 τ_1	0.0680(1)	127 ± 10	0.0680(1)	128 ± 10	0.999(1)	5.2
D3 ($\eta = 1.24$)	3–6 τ_1	0.07242(2)	3.7 ± 1.0	0.07244(4)	-0.5 ± 4	0.997(1)	13.5
	10–20 τ_1	0.072510(5)	5.2 ± 1.0	0.07250(1)	-0.4 ± 0.3	0.928(39)	
D4 ($\eta = 2$)	3–6 τ_1	0.07849(3)	0.1 ± 5	0.07847(2)	-0.8 ± 6	0.999(1)	–
	10–20 τ_1	0.0784700(6)	-0.03 ± 0.08	0.0784800(2)	0.00 ± 0.02	0.998(3)	
D9 ($\eta = 5.1$)	3–6 τ_1	0.104820(9)	0.5 ± 2	0.104870(6)	0.4 ± 1	1.000(1)	15
	10–20 τ_1	0.1048200(7)	0.03 ± 0.06	0.1048700(2)	0.00 ± 0.02	1.009(6)	

Table 6: The pole of h_1 in the PIC runs and in linear theory, with its uncertainty in the last digits, in parentheses (Section 2.5). R is the envelope ratio, PIC over linear, in the window. t_{nl} is set by the measurement floor in D1 and D2 (Section 5.4) and by the small frequency difference in D9, not by a dynamical departure; in D4 the residual never reaches 10 %.

On the damped side (D1, D2) the PIC runs and linear theory give the same frequency and damping rate, to 10^{-5} in ω and to 1 % in γ (Table 6), and the envelopes agree to 0.1 % while the

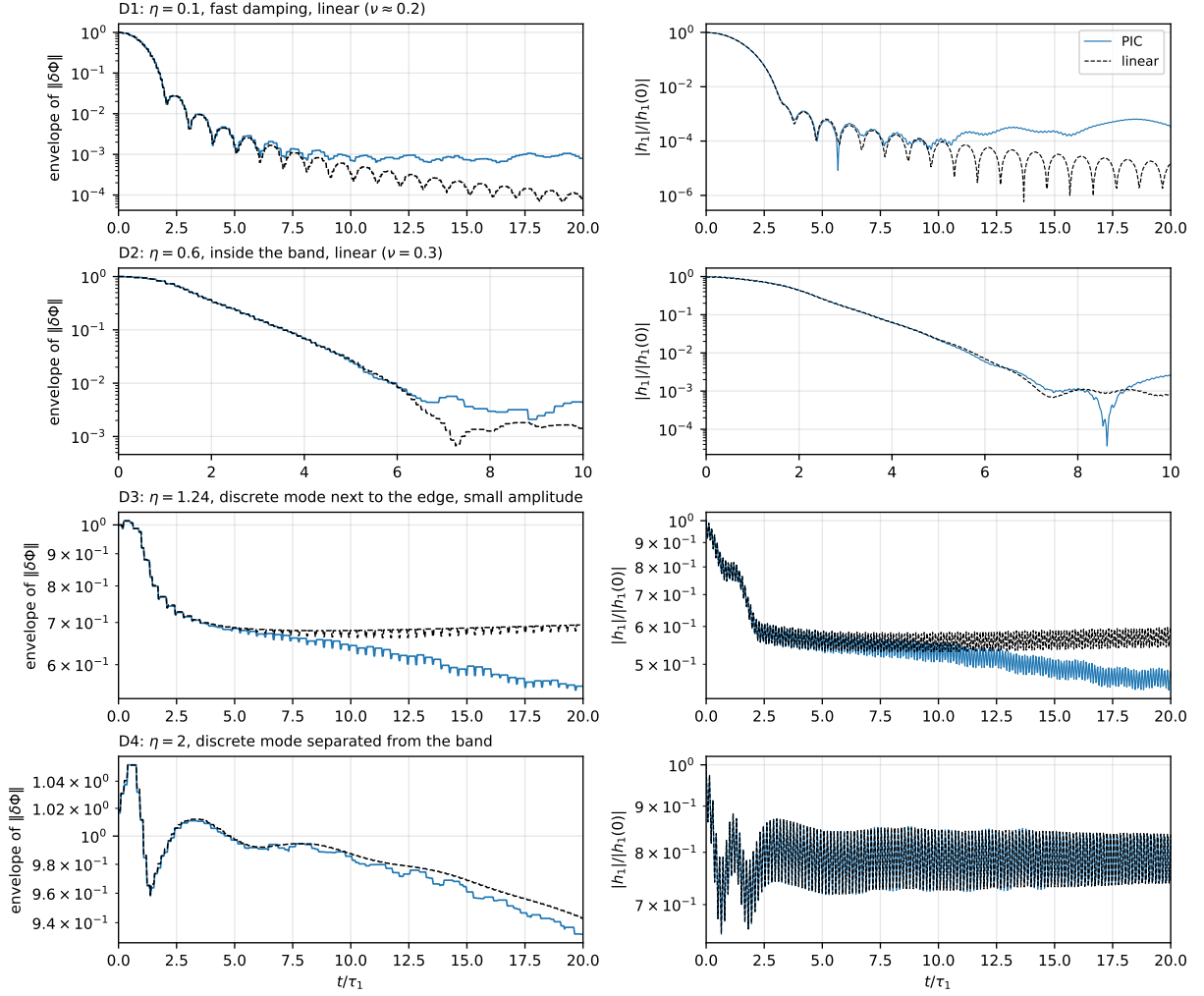


Figure 3: The four low-amplitude runs on both sides of the transition. Left: envelope of $\|\delta\Phi_\varepsilon\|$; right: $|h_1|$, both relative to their initial values. Blue, PIC; black dashed, linear theory for the same equilibrium and initial data. Time is in units of the τ_1 of each equilibrium.

signal decreases by two orders of magnitude. This agreement is much finer than the uncertainty of the pole itself, which over $1\text{--}3\tau_1$ is 3% in ω and 15% in γ for D1, and 10^{-4} and 8% for D2: the two signals are nearly identical, and what is poorly determined is the pole, not the comparison. D1 has $\varepsilon = 1$, so $\delta F/F_{\text{eq}}$ is of order one, and the evolution is nevertheless linear, because $\nu = 0.17$. The PIC runs depart from linear theory only when the signal reaches the measurement floor (Section 5.4): 6×10^{-4} in D1, at $3\tau_1$, and $\sim 2 \times 10^{-3}$ in D2, at $5\tau_1$.

On the discrete side (D3, D4) the potential perturbation keeps 55% to 100% of its initial value over $20\tau_1$, at the linear frequency to within 10^{-4} (relative). Self-gravity thus sustains the oscillation in the full simulation, not only in linear theory. The frequency difference, $1\text{--}5 \times 10^{-5}$ in absolute value (also in D9), is far above the fitting error ($< 10^{-6}$ over $10\text{--}20\tau_1$): the PIC runs give slightly slower discrete modes, a discretization effect of $\sim 10^{-4}$ (relative) that does not affect the conclusions. The two runs differ in how long the oscillation lasts. D4, with the mode well separated from the band ($x = -0.19$), loses 6.5% of its envelope between 5 and $20\tau_1$, and linear theory 5.4%; the excess is $\sim 1\%$, and the PIC–linear residual stays below 10% over the whole run. D3, with the mode just below the edge ($x = -0.01$), loses 19% over the same interval, which linear theory does not predict (Section 4.3).

4.2 Trapping near the transition

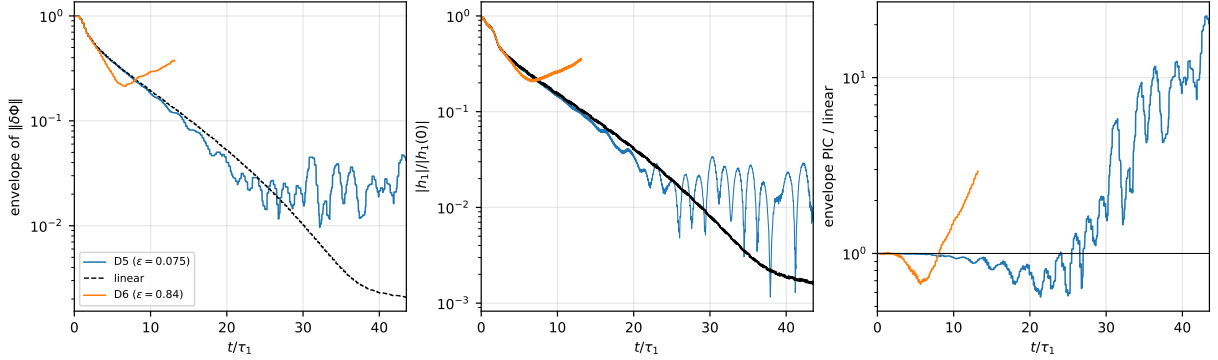


Figure 4: The same equilibrium, A4 ($\eta = 1$), at two amplitudes: D5 ($\varepsilon = 0.075$, $\nu = 3$) and D6 ($\varepsilon = 0.84$, $\nu = 10$). Dashed: linear theory, which is the same for both after division by ε . Right: envelope ratio, PIC over linear.

A4 lies at the edge of the transition: the linear pole is inside the band but very close to its edge ($x = 0.03$), and it is weakly damped, $\gamma = 3.7 \times 10^{-4}$. With $\varepsilon = 0.075$, $\delta F/F_{\text{eq}}$ never exceeds 7.5%, and any criterion based on ε would call the problem linear. O’Neil’s criterion gives $\nu = 3$. Figure 4 compares this run, D5, with D6, at $\varepsilon = 0.84$ and $\nu = 10$.

D5 follows linear theory for a long time. The pole of h_1 agrees up to $20\tau_1$ ($\gamma = 3.8 \times 10^{-4}$ against 3.7×10^{-4}), and the envelope ratio decreases slowly from 1 to 0.94. Between 20 and $30\tau_1$ the amplitude oscillates about the linear curve ($R = 0.76$), and from $\sim 25\tau_1$ on it stops decaying in the base run: over $30\text{--}44\tau_1$ its smooth part is 4.4 times the linear one. This late plateau is not converged (Section 5.3). It does not change with $\Delta t/2$, but with four times more particles the excess drops from 4.4 to 1.8 times the linear value and the fine part drops by a factor of eleven ($2.2 \times 10^{-2} \rightarrow 2.0 \times 10^{-3}$). Much of the plateau was discreteness noise, which in these runs grows with time, as documented in `PREGUNTAS_ABIERTAS.md`. Two resolutions cannot decide whether a physical excess remains; that would require $N \times 16$. The robust result of D5 is that it follows linear theory up to $\sim 20\tau_1$ at all three resolutions.

D6 departs from linear theory at $3\tau_1$. It first decays faster ($R = 0.81$ over $5\text{--}10\tau_1$). Then the damping stops ($\gamma = -4 \times 10^{-5}$ over $5\text{--}9\tau_1$, against $+3.4 \times 10^{-4}$ in linear theory) and the amplitude grows again, reaching twice the linear value at $13\tau_1$. The frequency increases by 0.6%

(0.0709 against 0.0704). All of this is converged: with four times more particles and with $\Delta t/2$, R is 0.801 and 0.805 over 5–10 τ_1 and 1.970 and 1.966 over 10–13 τ_1 , against 0.805 and 1.966 in the base run. The island is visible in (Q, J) (Figures 7 and 19): at the end, the rows with $J \gtrsim 0.11$ wind into a cat’s eye with an X point.

Comparison with the pendulum. To test the island quantitatively, the run D6L repeats D6 up to $t = 17\,200$ ($43\tau_1$), several bounce periods (Figure 5; `demo_eta.py isla`). The mode has $\omega = 0.07106$. Its resonance is at $J_r = 0.124$, where $|\Omega'(J_r)| = 0.095$, and the amplitude of the potential along that orbit, $A = |\delta\Phi_1(J_r)|$, averages 7.1×10^{-6} over 10–43 τ_1 . With these numbers, and no fit, the pendulum predicts two quantities.

- The half-width of the island, $2\sqrt{A/|\Omega'|} = 0.0173$. The measured value, the median over the trapped particles, is 0.0154 (maximum 0.025): the agreement is 10 %. The trapped particles are identified without reference to their J , as those whose phase $\psi = Q - \omega t$ librates. There are 1090 of them, all from the edge ($J_0 \geq 0.094$), carrying 0.2 % of the mass, because $F_{\text{eq}} \rightarrow 0$ there.
- The bounce period, $2\pi/\sqrt{A|\Omega'|} \approx 7700$. The amplitude of the mode oscillates with a period of ~ 9000 : 22–24 τ_1 between successive minima, at 6.6 and 30.2 τ_1 , and between maxima, at 18.0 and 40.1 τ_1 . This is 15 % to 20 % longer than the estimate, as expected, because particles near the separatrix bounce more slowly than those near the centre of the island. The second maximum is lower than the first (0.47 against 0.56), so the oscillation is damped. It is the amplitude oscillation that O’Neil’s theory predicts after trapping. With only two measured periods, this is a test at the 20 % level, not a precise fit.

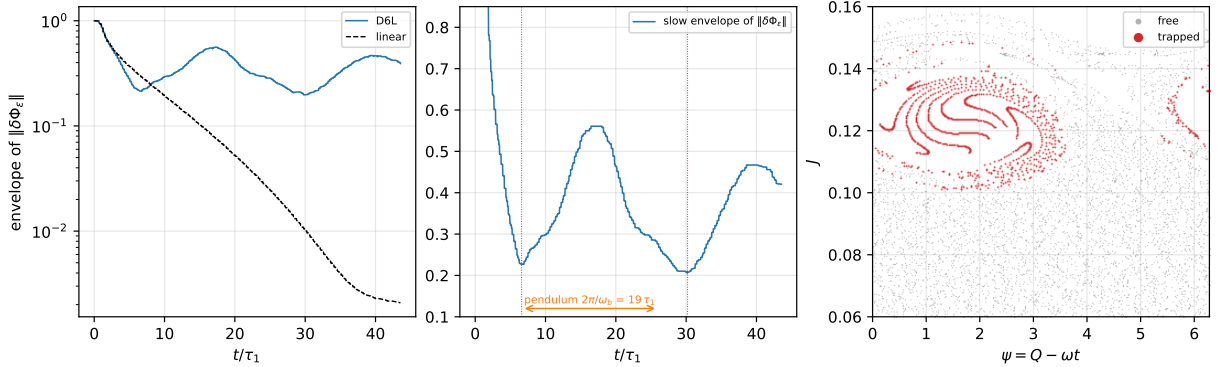


Figure 5: D6L, the run D6 continued to $43\tau_1$. Left: envelope of $\|\delta\Phi_\varepsilon\|$ and linear theory. Centre: slow envelope (window of ± 300); the amplitude oscillates with a period of ~ 9000 , and the arrow marks the bounce period of the pendulum with the measured amplitude. Right: the particles in the wave frame ($\psi = Q - \omega t$, J) at the end; in red, the trapped ones (their ψ librates over 10–43 τ_1), forming the cat’s eye of the island.

4.3 Discrete modes at finite amplitude

The loss of D3. In D3 the mode lies 2.5×10^{-4} below the edge ($x = -0.01$). The run loses 19 % between 5 and 20 τ_1 (from 0.685 to 0.555), while linear theory loses nothing (0.687 and 0.694). The fit of h_1 over 10–20 τ_1 gives $\gamma = (5.2 \pm 1.0) \times 10^{-5}$, against $(-0.4 \pm 0.3) \times 10^{-5}$ in linear theory, a difference of five times the error. Two independent checks support the fit. The envelope of the linear solution, without any fit, gives an effective rate between -6×10^{-6} and $+2 \times 10^{-6}$ over 5–20 τ_1 ; a linear γ of $4\text{--}5 \times 10^{-5}$ would make it lose 20 %, and it loses nothing. With synthetic signals of known γ at the same frequency, sampling and window, with a 5 % beat

and noise of 10^{-3} , the fit recovers γ to $\sim 10^{-6}$ between 0 and 10^{-4} (Section 2.5). The loss of D3 is therefore not linear. Nor is it numerical: with four times more particles, which doubles the resolution in J near the edge, it is 19.5 %, and with $\Delta t/2$ it is 19.8 %.

Mechanism. What distinguishes D3 is visible in the particles at the edge of the band, $J \approx J_t$. A discrete mode resonates with no particle of the support, but its resonance exists just outside it, at the action J_r where the orbital frequency, continued beyond the edge, equals ω :

$$J_r \approx J_t + \frac{\Omega_{\min} - \omega}{|\Omega'|}, \quad |\Omega'| \approx \frac{\Delta\Omega}{J_t}.$$

In L5 the mode is 2.5×10^{-4} below the edge and $J_r \approx J_t + 0.002 = 0.140$; in L6 it is 4.3×10^{-3} below, and $J_r \approx 0.164$. The equilibrium has no particles there, but the finite amplitude of the mode pushes the edge particles out of the support (Figure 7). In D3 the maximum action grows from 0.138 to 0.145 in $5\tau_1$ and to 0.151 at the end, beyond J_r , and the maximum change of action grows without saturating: 0.007, 0.013 and 0.020 at $t = 1800, 3600$ and 7200 . In D4 the maximum action reaches 0.142, far from its J_r , and the change of action stays at 0.003–0.004; in the unperturbed runs it is below 10^{-3} . We interpret this as follows: **the finite amplitude populates the resonance of a discrete mode lying just below the edge**, and the particles that reach it take energy from the mode, whereas a separated mode does not reach its resonance. In terms of scales, the bounce frequency of D3, $\omega_b \approx 1.5 \times 10^{-3}$, is six times the distance to the edge, while that of D4, 2.3×10^{-3} , is half of it. For a discrete mode, the ratio that decides linearity is $\omega_b/(\Omega_{\min} - \omega)$, not ν .

Amplitude scan. With the same equilibrium and four amplitudes, the loss tends to zero as ε decreases, the signature of a nonlinear effect (Figure 6 and Table 7). For $\varepsilon \leq 0.01$ the PIC rate agrees with the linear one within the error; the loss appears at $\varepsilon = 0.03$ and grows rapidly up to $\varepsilon = 0.1$ ($\gamma - \gamma_{\text{lin}} \propto \varepsilon^{1.7}$ between the two). The threshold lies where $\omega_b/(\Omega_{\min} - \omega)$ exceeds ~ 2 , and it coincides with the edge particles reaching the resonance. With $\varepsilon = 0.003$ they reach $J = 0.1393$, below $J_r \approx 0.140$; with $\varepsilon = 0.01$ they touch it (0.1410), and at larger amplitudes they go beyond it.

run	ε	$\frac{\omega_b}{\Omega_{\min} - \omega}$	loss	γ of h_1	$J_{\max}$	ΔJ_{edge}
D3e003	0.003	1.0	1.3 %	$(-2.8 \pm 0.9) \times 10^{-6}$	0.1393	0.0015
D3e01	0.01	1.9	3.1 %	$(-3.4 \pm 0.5) \times 10^{-6}$	0.1410	0.0031
D3e03	0.03	3.3	4.7 %	$(+3.3 \pm 0.6) \times 10^{-6}$	0.1454	0.0079
D3	0.1	6.0	19.8 %	$(+5.2 \pm 1.0) \times 10^{-5}$	0.1509	0.0199
D3N ($N \times 4$)	0.1	6.0	19.5 %	$(+1.4 \pm 3) \times 10^{-5}$	0.1510	0.0199
D3dt ($\Delta t/2$)	0.1	6.0	19.8 %	$(+5.2 \pm 1.0) \times 10^{-5}$	0.1509	0.0199

Table 7: The discrete mode of L5 at four amplitudes, and D3 at two further resolutions. The loss is that of the envelope of $\|\delta\Phi_\varepsilon\|$ between 5 and $20\tau_1$, net of the linear loss; γ is fitted over 10 – $20\tau_1$, and linear theory gives $\gamma = (-0.4 \pm 0.3) \times 10^{-5}$. $J_{\max}$ is the largest action reached at the end; the resonance of the mode is at $J_r \approx 0.140$. ΔJ_{edge} is the largest change of action of the edge rows ($J_0 > 0.12$). In D3N the fit of h_1 is noisier; the loss in $\delta\Phi$, which does not depend on it, agrees.

Large amplitude: D7. D7 takes the mode of D3 to $\varepsilon = 1$ (Figure 8). The mode is not damped: the envelope oscillates between 0.5 and 0.8 of its initial value over the $20\tau_1$, with a period of $\sim 14\tau_1$, instead of staying at 0.69 as in linear theory. The frequency increases by 0.4 %, from 0.0725 to 0.0728, and ends right at the edge of the band ($\Omega_{\min} = 0.0727$): at large amplitude

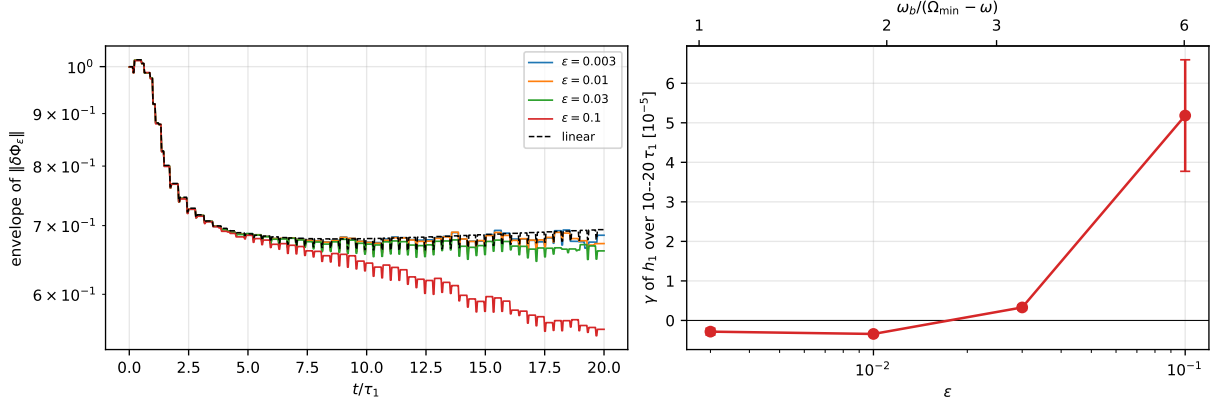


Figure 6: The ε scan of the mode of L5. Left: envelopes of $\|\delta\Phi_\varepsilon\|$ at four amplitudes and linear theory. Right: γ of h_1 over $10-20 \tau_1$ as a function of ε , with the ratio $\omega_b/(\Omega_{\min} - \omega)$ on the top axis.

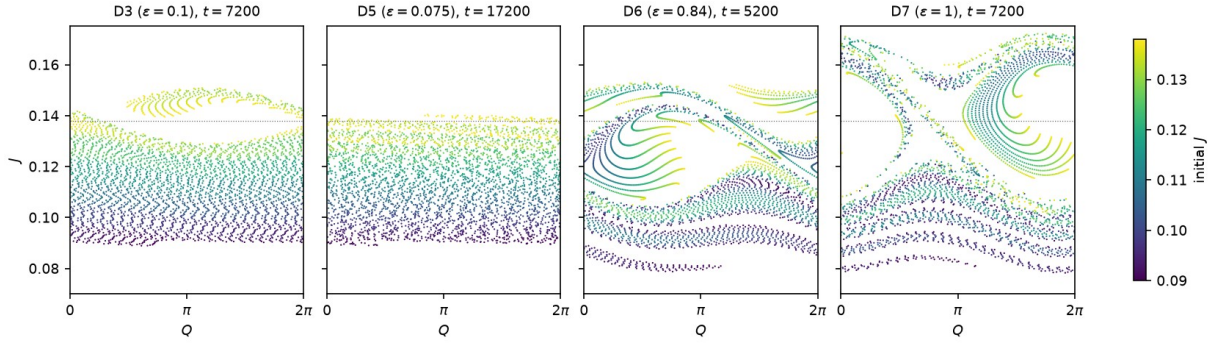


Figure 7: The edge of the band at the end of four runs: the particles with $J_0 > 0.09$ in (Q, J) , coloured by their *initial* action. Under free mixing each colour would stay in its horizontal row; the dotted line is the edge of the support, J_t . D3: the edge rows detach and form an arc outside the support, up to $J = 0.151$. D5: the change is small ($J \leq 0.140$). D6 and D7: islands of trapped particles, with their X point and closed curves around the centre, extending up to $J = 0.153$ and 0.179 . The position of the island in Q rotates with the wave and has no meaning.

the mode is no longer separated from the continuum. The particles change their action by up to 0.073, more than half the support, and reach $J = 0.179$, far outside it. The effective band extends downwards and the mode lies inside it, but it traps particles instead of being damped. Figure 7 shows the island with its X point. D3 and D7 together show that the mode survives at finite amplitude, but not as a linear mode: at small amplitude it slowly loses energy, and at large amplitude it reorganizes into an oscillating nonlinear structure.

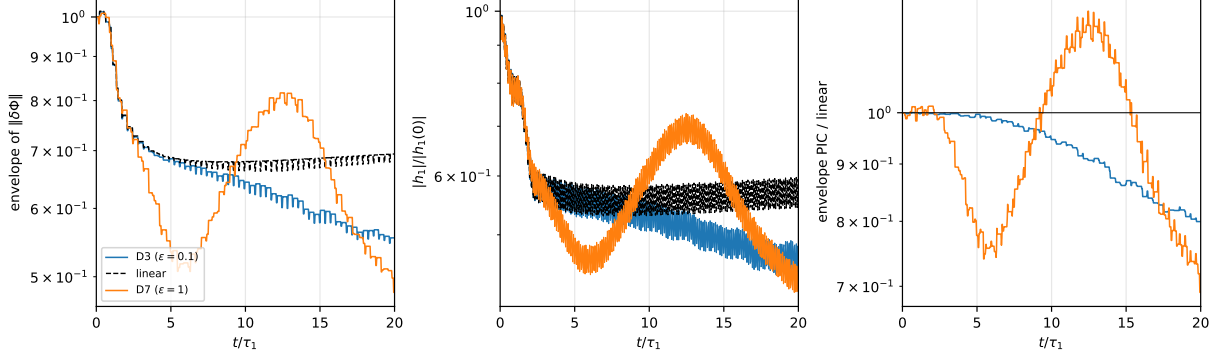


Figure 8: The discrete mode of L5 ($\eta = 1.24$) at two amplitudes: D3 ($\varepsilon = 0.1$) and D7 ($\varepsilon = 1$).

Mass comparable to the background: M5. In M5 the component has half the mass of the background. It contracts from $r_c = 5.7$ to 4.65, where the isochrone encloses a mass of 0.64: the component weighs almost as much as the background inside it, and its own field dominates the dynamics. Self-gravity speeds up the orbits by 150%: the band moves to $[0.134, 0.184]$ and τ_1 becomes 125. The linear mode is at $\omega = 0.10487$, 0.029 below the edge, more than half the width of the band ($x = -0.59$); the linear solution with half the time step gives the same value (0.10485). The perturbation is almost a pure normal mode: in linear theory the envelope of $\|\delta\Phi\|$ stays above 0.95 of its initial value over $40\tau_1$, without the mixing loss of D3 and D4.

The PIC runs reproduce it at both amplitudes (Figure 9 and Table 6). The frequency is 0.10482 with $\varepsilon = 0.1$ and 0.10485 with $\varepsilon = 1$, against 0.10487 in linear theory (5×10^{-4} and 2×10^{-4} relative), and the fit finds no damping: $|\gamma| < 10^{-6}$ over 10–20 and 20–40 τ_1 . Projected on the mode profile, the PIC amplitude agrees with the linear one to 1% up to $10\tau_1$; afterwards the ratio oscillates between 0.90 and 1.08 ($\varepsilon = 0.1$) and between 0.94 and 1.05 ($\varepsilon = 1$) because of the beat. The value $t_{nl} = 15\tau_1$ in Table 6 is not a nonlinear effect: the small frequency difference accumulates a quarter of a radian of phase over $40\tau_1$, and the PIC–linear residual grows with it.

The difference from D3 and D7 is the distance to the resonance. For the mode of M5 it lies at $J_r \approx J_t + 0.029/|\Omega'| \approx 0.22$, 0.08 beyond the edge of the support. With $\varepsilon = 0.1$ the edge particles reach $J = 0.140$, with a maximum change of action of 0.003 that hardly grows after $10\tau_1$. With $\varepsilon = 1$ they reach 0.157, with a change of 0.02, also saturated after $10\tau_1$. In neither case do they approach J_r . The ratio $\omega_b/(\Omega_{\min} - \omega)$ is 0.2 with $\varepsilon = 0.1$ and 0.63 with $\varepsilon = 1$, below one in both cases. These two runs were the first test of prediction P3 made before the run, and the mode survives as predicted. With $\varepsilon = 1$ the static second-order part is larger (7×10^{-3} of the initial value, against $\sim 10^{-3}$ with $\varepsilon = 0.1$), of the size expected for an effect $\propto \varepsilon^2$.

4.4 The vacuum edge

D1 and D8 have the same relative mass ($\eta = 0.1$) and differ only in the exponent of the vacuum edge, $g = 2$ and $g = 1$ (Figure 10). Up to $2\tau_1$ they are nearly identical. Afterwards the run with the sharp edge decays much more slowly: over 10–20 τ_1 its envelope is 2.6×10^{-3} , five times the linear envelope of D1. The PIC run follows the linear theory of D8 to 6% up to $\sim 15\tau_1$, and the late pole agrees: $\omega = 0.06088 \pm 0.00005$ against 0.06095 ± 0.00003 over 10–20 τ_1 , just below

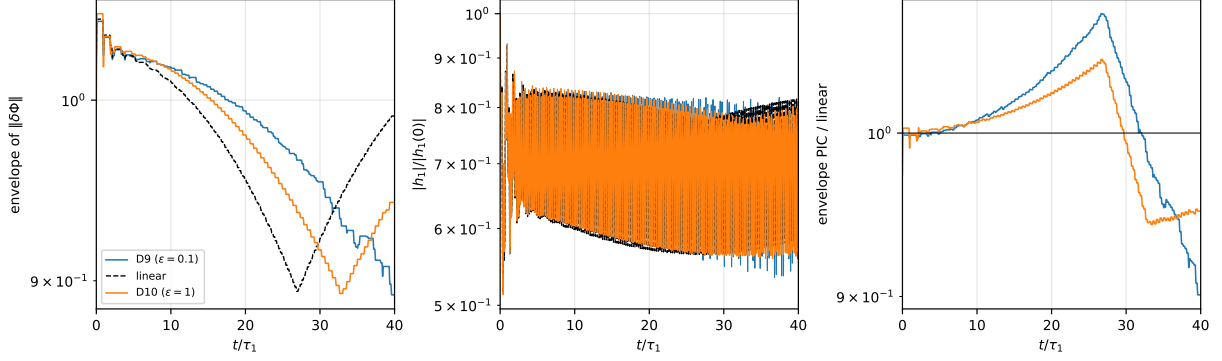


Figure 9: M5 ($a_0 = 0.5$, $\eta = 5.1$) at two amplitudes: D9 ($\varepsilon = 0.1$) and D10 ($\varepsilon = 1$). The left vertical axis spans only 0.9 to 1.05: the envelope is modulated by about $\pm 7\%$ by a slow beat that linear theory also has. In the PIC runs the beat is slower, which makes the ratio on the right oscillate; there is no damping.

the edge of the band ($\Omega_{\min} = 0.06101$). This is the edge response that persists in linear theory for $g \leq 1$, the dichotomy of Hadžić et al. in the full simulation. Beyond $15\tau_1$ the signal of D8 reaches the floor of the runs with $\varepsilon = 1$, and the PIC run no longer distinguishes the two edges.

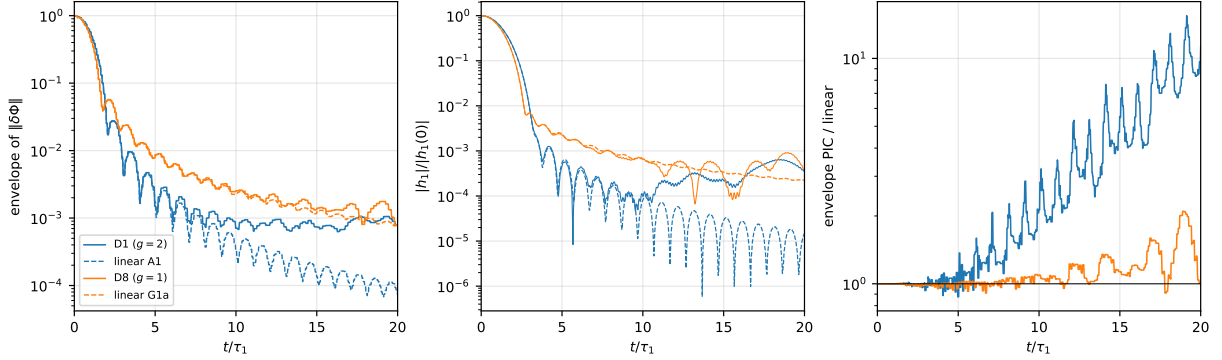


Figure 10: The same $\eta = 0.1$ with a smooth edge (D1, $g = 2$) and a sharp edge (D8, $g = 1$). Left and centre: envelope of $\|\delta\Phi\|$ and $|h_1|$; dashed, the linear theory of each equilibrium. Right: envelope ratio, PIC over linear. D1 reaches the measurement floor near $5\tau_1$ (Section 5.4); D8 stays above it until $\sim 15\tau_1$.

4.5 The blob

There is a transition in the mass, and it is ordered by μ (Figure 11 and Table 8).

- With $a_0 = 10^{-3}$ ($\mu = 0.1$) the blob mixes almost as it does without self-gravity: $|h_1|$ drops by four orders of magnitude in $10\tau_1$.
- With $a_0 = 10^{-2}$ ($\mu = 0.3$), the largest mass of the earlier runs, it also mixes, but $|h_1|$ levels off at 3×10^{-3} , forty times above free mixing, and a remnant core stays in (r, p_r) (Figure 24).
- With $a_0 = 0.03$ ($\mu = 0.5$) part of the blob mixes and part does not: $|h_1|$ drops to 0.2 and stays there, with 40% of the mass in a core.
- With $a_0 = 0.1$ and 0.3 ($\mu \approx 0.9$ and 1.2) the blob does not disperse. It collapses in the first radial periods into a dense core with 79% and 99% of the mass, and $|h_1|$ stays at 0.7–0.8 during the $31\tau_1$.

run	a_0	N_{pc}	η	μ	$\langle h_1 \rangle$ over $10\text{--}20\tau_1$		
					instantaneous	bare	core
B0	–	25	0	0	7.0×10^{-5}	7.0×10^{-5}	–
B1	10^{-3}	25	0.005	0.10	1.1×10^{-4}	1.0×10^{-3}	–
B2	10^{-2}	25	0.05	0.31	2.6×10^{-3}	1.1×10^{-2}	–
B3	0.03	25	0.16	0.52	0.22	0.22	40 %
B4	0.1	25	0.60	0.89	0.80	0.81	79 %
B5	0.3	25	2.5	1.16	0.69	0.83	99 %
B4q	0.1	100	0.60	0.89	0.85	0.86	79 %

Table 8: The blob at different masses. B0 has no self-gravity. η is that of the distribution averaged over Q ($J^2 e^{-J^2/0.1^2}$, `eta.py`); $\mu = \omega_b/\Delta\Omega$, with ω_b estimated from the first radial period. $\langle |h_1| \rangle$ is computed with the map of the instantaneous potential and with that of the bare isochrone. “Core” is the fraction of the mass with $J < 0.06$ at the end, in the instantaneous potential. It is meaningful only when there is a core (B3–B5): at small mass this cut also collects the low end of the distribution.

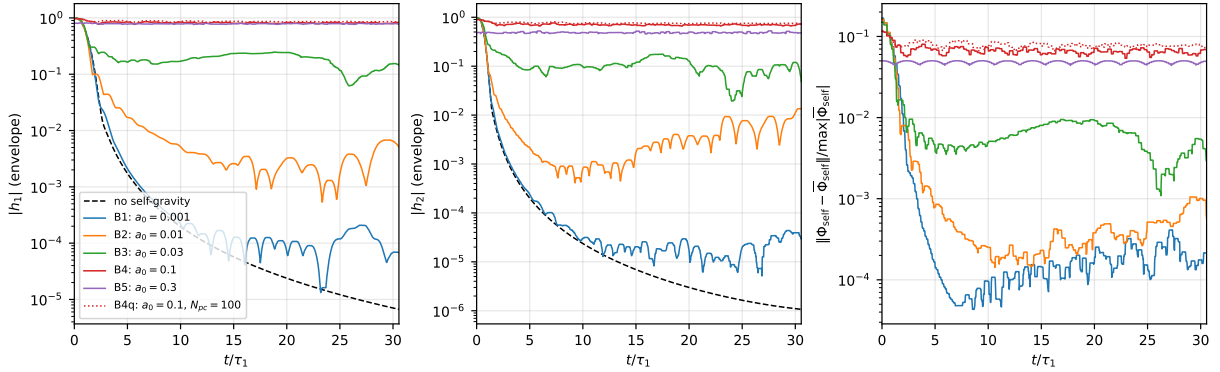


Figure 11: The blob at different masses. Left and centre: envelopes of $|h_1|$ and $|h_2|$ in the map of the instantaneous potential; dashed, without self-gravity. Right: fluctuation of the self-potential about its late-time mean. Between $a_0 = 10^{-2}$ and 0.1 the behaviour changes from mixing to a blob that does not disperse.

The transition lies between $\mu \approx 0.3$ and 0.5 . Above it, the well of the blob traps its particles before the shear separates them.

In this radial model a blob in (r, p_r) is a thin spherical shell. Its own potential has a kink at the radius of the shell, which creates a local well. The particles are trapped in it, and the shell moves as a single object: in B4 its mean radius oscillates by ± 0.8 around 5.7 , with frequency 0.075 , above the band of the isochrone. In the map of the instantaneous potential the core appears at $J \approx 0$, since it is the nearly circular orbit of the potential that it creates. Around it remains a halo of particles stripped in the first radial periods, and these do mix (Figure 25). This is the extreme case of the islands of D6 and D7: there the wave trapped a fraction of the particles, and here it traps almost all of them.

The observation from the earlier runs, that at larger mass a blob attracts the particles and the h_k stop tending to zero, is correct, and the mechanism is nonlinear: self-trapping, controlled by μ . It is not the linear discrete mode of D3–D10. For this distribution η is 0.005 , 0.05 , 0.16 , 0.60 and 2.5 from B1 to B5, and the transition occurs between B2 and B3, at $\eta \approx 0.05$ – 0.16 , far below $\eta_c \approx 1.1$, where linear theory predicts mixing. At the masses of the earlier runs ($a_0 \leq 10^{-2}$), however, the effect is small and must be measured with care.

- With the map of the bare isochrone, the one the code uses, $|h_1|$ levels off at 10^{-3} ($a_0 = 10^{-3}$) and 10^{-2} ($a_0 = 10^{-2}$). With the map of the actual potential it drops to 10^{-4} and 3×10^{-3} . The difference is a coordinate effect, already documented in `PREGUNTAS_ABIERTAS.md`: measuring orbits that live in another potential with the isochrone map leaves a plateau that is not physical.
- In B2 the high harmonics grow at late times ($|h_4| \approx 3 \times 10^{-2}$ over 20 – $31 \tau_1$). This is consistent with the growth of discreteness noise that `PREGUNTAS_ABIERTAS.md` measures for $a_0 = 10^{-2}$ ($\propto t^{1.9}$, smaller for larger N_{rc}), and we do not interpret it as physical.

B4q, with four times more particles in angle, gives the same core mass (79%) and a late $|h_1|$ larger by 6% (0.85 against 0.80): the conclusion does not depend on the resolution in Q , although the exact value does slightly. The resolution in J and the time step have not been varied. The open points are a finer mass scan between 0.01 and 0.1 , the dependence of the critical μ on the angular width s_Q of the blob (a wider blob has less self-potential per unit mass), and convergence in N_{rc} and Δt .

5 Validation

5.1 Energy conservation

The code stores the total energy at each snapshot. In the 17 runs, $\max_t |E(t) - E(0)|/|E(0)| \leq 4 \times 10^{-6}$, with no appreciable drift, and in the 14 control runs (the ε scan, the convergence runs and D6L) it stays below 10^{-6} (Figure 12; full table in `exe/demo_eta/energia.txt`). In the blob runs it reaches 1.7×10^{-4} in B5, the most violent collapse, and 4×10^{-5} in B4.

5.2 Stationarity of the equilibria

The seven Z runs have the same equilibrium and initial positions as their D runs, without perturbation. In them the rows of constant J stay horizontal (Appendix B.1): the largest change of action of any particle over the run ranges from $1.2 \times 10^{-4} J_t$ (Z_A1) to $9.5 \times 10^{-3} J_t$ (Z_M5). To find what sets these changes we use four measures, computed every 50 time units: the largest change of action, $\max_i |J_i(t) - J_i(0)|$; the mean change weighted by mass, $\langle |\Delta J| \rangle = \sum_i w_i |J_i(t) - J_i(0)| / \sum_i w_i$; the rms of $\delta\Phi$ over $3 \leq r \leq 15$, relative to $\max |\Phi_{\text{self}}|$; and the energy error. The actions are computed with the angle–action map of the equilibrium and are given in units of J_t .

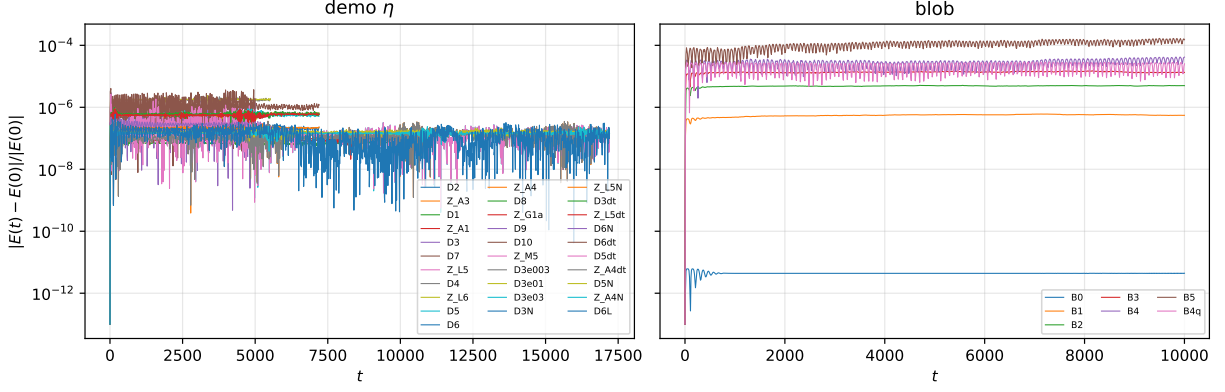


Figure 12: Conservation of the total energy. Left: the 17 runs and the 14 control runs. Right: the blob runs; B0 has no self-gravity. All runs use yoshida4 with $\Delta t = 0.05$, except the $\Delta t/2$ controls.

The two constructions. The equilibria of A1, G1a, A4 and M5 were also built with the differential equation (Section 2.1), loaded on the same nodes and evolved with the parameters of the Z runs (runs E, `equilibrio_edo_pic.py`). Table 9 compares the two. All the measures agree to better than 1%. The difference between the two equilibria, $\sim 10^{-6}$, is not visible in the PIC code, and the departures from stationarity come from the PIC discretization.

case	equilibrium	t	$\max_i \Delta J $	$\langle \Delta J \rangle$	$\delta\Phi(0)$	$\overline{\delta\Phi}$	$ \Delta E/E $
A1	ODE	5550	6.14×10^{-5}	8.81×10^{-6}	5.76×10^{-5}	5.72×10^{-5}	1.20×10^{-7}
	iteration		6.15×10^{-5}	8.83×10^{-6}	5.79×10^{-5}	5.75×10^{-5}	1.20×10^{-7}
G1a	ODE	5550	6.25×10^{-5}	9.49×10^{-6}	5.52×10^{-5}	5.48×10^{-5}	1.04×10^{-7}
	iteration		6.26×10^{-5}	9.51×10^{-6}	5.55×10^{-5}	5.52×10^{-5}	1.04×10^{-7}
A4	ODE	3950	1.75×10^{-3}	1.30×10^{-4}	6.12×10^{-5}	6.99×10^{-5}	2.81×10^{-7}
	iteration		1.76×10^{-3}	1.30×10^{-4}	6.15×10^{-5}	7.02×10^{-5}	2.81×10^{-7}
M5	ODE	5000	9.18×10^{-3}	1.28×10^{-3}	8.69×10^{-5}	1.80×10^{-4}	1.25×10^{-6}
	iteration		9.26×10^{-3}	1.29×10^{-3}	8.73×10^{-5}	1.80×10^{-4}	1.25×10^{-6}

Table 9: Evolution at $\varepsilon = 0$ of the equilibria of the differential equation (runs E) and of the iteration (runs Z), at the resolution of the demo. The changes of action are at time t , in units of J_t . $\delta\Phi(0)$ is the rms of $\delta\Phi$ at $t = 0$ and $\overline{\delta\Phi}$ its mean over the second half of the interval, both relative to $\max |\Phi_{\text{self}}|$. The last column is the largest energy error up to t .

The error of the PIC field. The resolution runs of M5 in Table 10 separate two effects. They repeat E_M5 with four times more particles, with half the cell ($\Delta r = 0.05$ with the same Δt , so the Courant factor is $C = 2$), with both, and with half the cell and half the time step.

The first effect is deterministic. At $t = 0$ the potential that the code computes from the particles differs from that of the equilibrium by the error of the deposition and of the field solve. In M5 the rms difference is 8.7×10^{-5} . The particles sit on orbits of the exact potential, so in the potential of the code they are slightly out of equilibrium, and the system adjusts. After a few radial periods $\delta\Phi$ has a static part, with rms 1.7×10^{-4} , and an oscillation at the frequency of the discrete mode of M5, $\omega = 0.1048$ (Section 4.3). The oscillation has a constant rms amplitude of 7.7×10^{-5} from $t = 500$ to 5000: the mismatch excites the mode, and the mode does not damp. These three numbers do not depend on N or Δt , to 3%, and they decrease by a factor of 3.9 to 4.1 when Δr is halved. They are the $O(\Delta r^2)$ error of the PIC field. The same happens in the other equilibria with a discrete mode: in Z_L5 and Z_L6, $\delta\Phi$ oscillates at the frequencies of the modes of D3 and D4 (Table 6), with rms amplitudes of 8.9×10^{-6} and 2.3×10^{-5} . Without a discrete mode (A1, A3, A4 and G1a) the oscillation is below 4×10^{-6} , and only the static part

remains, $5.5\text{--}7.0 \times 10^{-5}$.

The actions respond to the same error. On an orbit of the code, the energy in the potential of the equilibrium, $E - \delta\Phi(r)$, varies by $\delta\Phi$ along the orbit, and since $dJ/dE = 1/\Omega$ the action measured with the map of the equilibrium varies by about $|\delta\Phi|/\Omega$. With $\delta\Phi$ of Table 9 and Ω at the centre of the band, this estimate is $6.9 \times 10^{-6} J_t$ in A1, $8.7 \times 10^{-5} J_t$ in A4 and $8.7 \times 10^{-4} J_t$ in M5. It is within a factor of 1.5 of the measured $\langle |\Delta J| \rangle$, over a range of more than two orders of magnitude. In M5 the measured value depends little on N (1.28 and $1.21 \times 10^{-3} J_t$ at $\Delta r = 0.1$), and with $\Delta r/2$ and four times more particles it is $3.2 \times 10^{-4} J_t$, 3.8 times less. In E_M5, E_M5N and E_M5drN, $\langle |\Delta J| \rangle$ grows by only 20–30 % between $t = 1000$ and 5000.

run	Δr	particles	Δt	$\langle \Delta J \rangle [10^{-3} J_t]$		$\delta\Phi [10^{-5}]$			$ \Delta E/E $
				$t = 1000$	5000	$t = 0$	static	mode	
E_M5	0.1	400×25	0.05	1.01	1.28	8.69	17.2	7.66	1.3×10^{-6}
E_M5N	0.1	800×50	0.05	0.99	1.21	8.63	17.2	7.63	1.1×10^{-6}
E_M5dr	0.05	400×25	0.05	0.85	4.22	2.15	4.28	1.98	2.3×10^{-6}
E_M5drdt	0.05	400×25	0.025	0.85	4.18	2.15	4.26	1.98	2.3×10^{-6}
E_M5drN	0.05	800×50	0.05	0.25	0.32	2.13	4.27	1.92	2.1×10^{-7}

Table 10: Resolution runs of M5 at $\varepsilon = 0$, from the equilibrium of the differential equation. With 800×50 particles the resolution in J is also doubled. $\delta\Phi$ is the rms over $3 \leq r \leq 15$ relative to $\max |\Phi_{\text{self}}|$: at $t = 0$; of its time average over $500 \leq t \leq 5000$ (static); and of the rest (mode), which oscillates at the frequency of the discrete mode. The last column is the largest energy error.

Discreteness of the particles. The second effect is a growth of the spread of the actions. It is clear with $\Delta r/2$ and 400×25 particles (E_M5dr): $\langle |\Delta J| \rangle$ grows almost linearly in time, from $8.5 \times 10^{-4} J_t$ at $t = 1000$ to $4.2 \times 10^{-3} J_t$ at $t = 5000$, and the largest change reaches $4.7 \times 10^{-2} J_t$. With half the time step the result is the same (E_M5drdt), so the time integration is not the cause. With four times more particles the growth disappears (E_M5drN). The spread is symmetric: in E_M5dr the mean change of action, weighted by mass, is $3.5 \times 10^{-7} J_t$, 10^{-4} of $\langle |\Delta J| \rangle$, so there is no heating. $\delta\Phi$ is also the same as with four times more particles. The energy error behaves in the same way: it is 2.3×10^{-6} in E_M5dr and E_M5drdt, and 2.1×10^{-7} with four times more particles. The same growth is present at $\Delta r = 0.1$, but it is much slower. In Z_A4, $\langle |\Delta J| \rangle$ rises from $1.0 \times 10^{-4} J_t$ at $t = 1000$ to $2.9 \times 10^{-4} J_t$ at $t = 17200$. With $\Delta t/2$ (Z_A4dt) the values are the same to three digits, and with four times more particles (Z_A4N) $\langle |\Delta J| \rangle$ is 9.2 and $8.8 \times 10^{-5} J_t$ at the same times. L5 behaves in the same way. As for the fine part of $\delta\Phi$ (Section 5.4), the growth falls with N much faster than Monte Carlo noise would, as expected with a quiet start, whose noise is not Monte Carlo noise. At fixed N a finer cell makes the growth faster, so refining Δr requires more particles.

At the resolution of the demo the error of the PIC field dominates. It is almost the same in a D run and in its Z run, which have the same particles, so it cancels in $\delta\Phi_\varepsilon$ up to the floors of Section 5.4. The growth from discreteness adds about $2 \times 10^{-4} J_t$ to $\langle |\Delta J| \rangle$ by $t = 17200$ in Z_A4. Neither effect depends on how the equilibrium was built.

5.3 Convergence in N and Δt

D3, D5 and D6 were repeated with four times more particles ($400 \times 25 \rightarrow 800 \times 50$, which also doubles the resolution in J) and with $\Delta t/2$, each with its own reference run at the same resolution (Figure 13).

- D3 is converged. The loss is 19.8 %, 19.5 % and 19.8 % at the three resolutions, and the smooth part relative to the linear one over $10\text{--}20 \tau_1$ is 0.93 in all three.

- D6 is converged: $R = 0.805, 0.801$ and 0.805 over $5\text{--}10\tau_1$, and $1.966, 1.970$ and 1.966 over $10\text{--}13\tau_1$. Its fine part does not change with N , so it is structure of the mode, not noise.
- D5 is converged in Δt but not in N after $20\tau_1$. Over $30\text{--}43.5\tau_1$ the smooth part is $4.4, 1.8$ and 4.4 times the linear one, and the fine part is $2.2 \times 10^{-2}, 2.0 \times 10^{-3}$ and 2.2×10^{-2} . Up to $20\tau_1$ the three agree ($R = 0.94\text{--}0.96$).

The tables are in `exe/demo_eta/controles.txt` (`demo_eta.py controles`).

The smooth/fine split does not depend on the smoothing scale. With Gaussian widths of $0.25, 0.5$ and 1.0 in r , the PIC-to-linear ratio of the smooth part of D5 over $30\text{--}44\tau_1$ is $4.44, 4.40$ and 4.31 , and the projection on the mode profile, which uses no scale, gives 4.45 . The fine part is $\sim 2 \times 10^{-2}$ with all three widths. In D6 the ratio is 2.0 with all three.

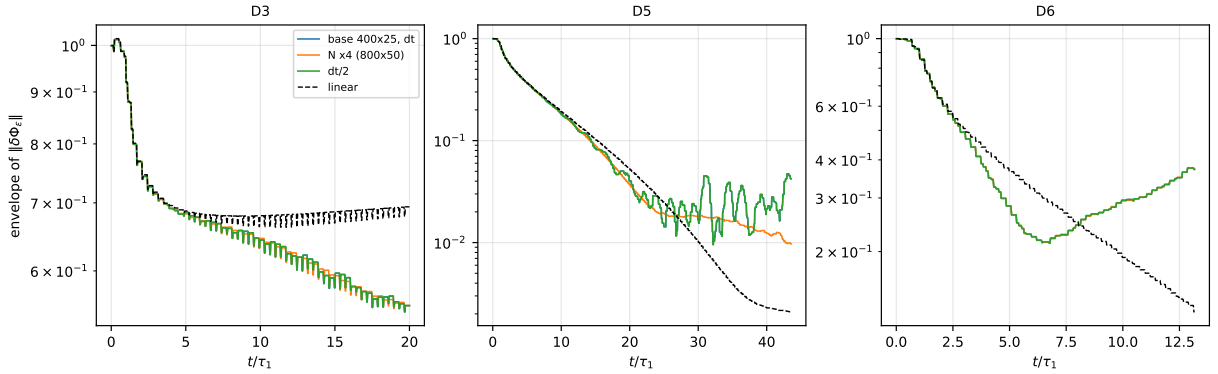


Figure 13: Convergence of D3, D5 and D6: envelope of $\|\delta\Phi_\varepsilon\|$ at the resolution of the runs (400×25 , $\Delta t = 0.05$), with four times more particles (800×50) and with $\Delta t/2$; dashed, linear theory. The $\Delta t/2$ curve lies on top of the base curve in all three cases. In D5 the $N \times 4$ curve separates from the other two after $\sim 20\tau_1$.

5.4 Measurement floors

Four effects limit how far a decaying signal can be followed.

- **Discretization.** At the start, PIC and linear theory differ by $\sim 10^{-3}$ of the initial value of $\|\delta\Phi\|$ in all runs. This is the combined error of the two discretizations.
- **Static second-order part.** The PIC–linear residual has a part that does not oscillate and is constant after $\sim 3\tau_1$: 5.7×10^{-4} of the initial value in D1 and 7×10^{-4} in D8. It is physical. After division by ε it grows as ε (between D6 and D5 the ratio is $10\text{--}11$ depending on the window, and that of the amplitudes is 11.2), so it is an effect $\propto \varepsilon^2$: the change of the mean radial profile ($k = 0$) that the perturbation leaves as it mixes. Since it does not oscillate it can be subtracted, and what remains is $\sim 2 \times 10^{-4}$.
- **Subtraction noise.** The Z run is not exactly stationary: its $\delta\Phi$ has a static part and, in L5, L6 and M5, an oscillation of the discrete mode, both caused by the $O(\Delta r^2)$ error of the PIC field (Section 5.2), and fluctuations. The subtraction cancels them almost completely, but the remainder is divided by ε . With $\varepsilon \approx 0.07$ the floor is $\sim 2 \times 10^{-3}$ (D2).
- **Fine structure.** The part of $\delta\Phi_\varepsilon(r)$ on the scale of a few grid cells, which in the linear solution is a fixed fraction of the signal, has a floor in the PIC runs: 4×10^{-4} in D1 and D8 ($\varepsilon = 1$) and 4×10^{-3} in D2 ($\varepsilon = 0.065$), again $\propto 1/\varepsilon$. In D5 it grows with time, up to 2×10^{-2} at $44\tau_1$, and it is noise: with four times more particles it drops by a factor of eleven. This is much faster than $N^{-1/2}$, because with a quiet start the noise is not Monte

Carlo noise. For long tails it is better to measure the smooth part, or h_1 , which does not go through the grid, and to check with larger N .

Since the second effect grows with ε and the third decreases, there is an optimal amplitude for following long tails, of order $\varepsilon \approx 0.3\text{--}0.5$, with a floor of about 5×10^{-4} . This is an estimate from these runs, not a measurement.

Correction to the previous version. A previous version of the comparison paired each PIC snapshot with a sample of the linear solution that, because of rounding accumulated in the sample times, was in $\sim 60\%$ of the cases the next one, two time units later. This does not affect the envelopes, the ratios R or the PIC poles, but it does affect the PIC–linear residuals. With the corrected pairing, t_{nl} of D3 changes from 7.4 to $13.5 \tau_1$ and that of D5 from 6.7 to $11.7 \tau_1$, D4 never reaches it, and the excess loss of D4 over linear theory drops from 5% to 1%. All numbers in this document use the corrected pairing.

6 Discussion and conclusions

6.1 Limitations of the fixed- L model

All of the above holds for the reduction $f = F(r, p_r) \delta(L - L_0)$, which is an extremely anisotropic model. Of a real spherical system it retains the radial dynamics and the edge Ω_{min} of the band. It loses three things.

- **There is no spread in L .** In a system with distributed L , mixing involves two frequencies (radial and azimuthal), and the spread in L blurs the edge $\Omega_{\text{max}} = \kappa$. Here that edge is exact and artificial.
- **Only radial perturbations exist ($\ell = 0$).** There are no non-spherical modes and no resonances between the radial and azimuthal frequencies.
- **The orbital structure is rigid.** The measure $8\pi^2 L_0 dr dp_r$ fixes the weight of each orbit, and the breathing mode is the only collective degree of freedom.

These limitations apply relative to a real spherical system. Relative to the theorem of Hadžić et al. there is no such gap: it is formulated in the same fixed- $|L|$ reduction, with a point mass instead of the isochrone as background, so the comparison is direct (Section 1.6). This work addresses what the theorem leaves open: the mass up to which damping holds for $g > 1$ (the $\varepsilon_0(k)$ of the theorem, estimated here as $\eta_c \approx 1.1\text{--}1.2$ for $g = 2$), what happens above it, and the nonlinear dynamics.

6.2 Conclusions

Demonstrated (with the controls of Section 5):

- The linear solver has a transition from a damped mode to a discrete mode at $\eta_c \approx 1.1\text{--}1.2$ for this family.
- The PIC runs reproduce linear theory where $\nu \ll 1$, to 10^{-5} in ω and 1% in γ , and they reproduce the discrete modes over $20\text{--}40 \tau_1$.
- Near η_c , with $\nu = 10$ (D6), the PIC run departs from linear theory: damping stops and the amplitude oscillates. This is converged in N and Δt .
- The discrete mode just below the edge (D3) loses amplitude while linear theory does not ($\gamma_{\text{PIC}} = (5.2 \pm 1.0) \times 10^{-5}$, $|\gamma_{\text{lin}}| \lesssim 6 \times 10^{-6}$). The loss is converged in N and Δt and vanishes as ε decreases, so it is nonlinear.

- A blob in phase with $a_0 \gtrsim 0.03$ does not disperse.
- The three linear regimes are those of the theorem of Hadžić et al., formulated in the same reduction. With a smooth edge ($g = 2$) and small mass the perturbation is damped. With a King edge ($g = 1$) and small mass there is an oscillation just below $\Omega_{\min}$ (D8: $\omega = 0.06088 \pm 0.00005$ against $\Omega_{\min} = 0.06101$). With $g = 2$, damping ceases above $\eta_c \approx 1.1$ – 1.2 , a mass that the theorem does not estimate.

Interpreted (supported by the particle data and by quantitative tests at the 10 % to 20 % level):

- The saturation of D6 is trapping of the O’Neil type: the half-width of the island agrees with the pendulum to 10 %, and the amplitude oscillates with its bounce period to 20 %, with only two measured periods.
- The loss of D3 is caused by the finite amplitude pushing edge particles to the resonance of the mode, outside the support: the threshold in ε coincides with those particles reaching J_r .
- The blob is held together by self-trapping, ordered by $\mu = \omega_b/\Delta\Omega$.

Hypotheses (not tested here):

- That the late excess of D5 over linear theory is physical. With four times more particles it drops from 4.4 to 1.8 times the linear value, and it is not converged.
- That η_c is universal. Linear theory shows that it depends on the width, the shape and the edge of the distribution.
- That the criterion $\omega_b/(\Omega_{\min} - \omega)$ holds in general for discrete modes.
- That η_c for $g = 2$ coincides with the $\varepsilon_0(2)$ of the theorem of Hadžić et al. Here the background is the isochrone, not a point mass; with `BGtype = sphere` the code can reproduce their case.
- That any of this carries over to spherical systems with distributed L (Section 6.1).

Outcome of the predictions. P1 holds in D1, with $\varepsilon = 1$, and in D2. P2 holds in D6, converged, with the bounce period of the pendulum to 20 %; in D5 the late departure is not converged in N and counts neither for nor against it. P3 holds blind in D9 and D10, where the ratio is 0.2 and 0.63 and the mode survives, and the ε scan of D3 confirms it from the other side: the loss disappears when the ratio falls below ~ 2 . For the blob a transition near $\mu \sim 1$ was expected; it appears between $\mu = 0.3$ and 0.5 .

6.3 Next steps

1. **D5 at late times.** With two resolutions it is not known whether a physical excess over linear theory remains after $20\tau_1$. This requires $N \times 16$ (1600×100 , about 12 hours with its reference run) or a diagnostic that does not depend on the grid.
2. **The full (η, ν) map** on the damped side, at fixed ε , to see where the departure starts as a function of ν ; and D6 at two or three amplitudes, to test whether the width of the island scales as $\sqrt{\varepsilon}$.
3. **The sharp edge at late times**, with $\varepsilon \approx 0.3$ – 0.5 , the amplitude that minimizes the floor, and with the static part subtracted, to follow the response of D8 beyond $15\tau_1$.

4. **The case of the theorem as stated**, with a point mass as background (BGtype = **sphere**), to estimate $\varepsilon_0(k)$ in the same problem as Hadžić et al.

The complete design, with 30 runs and their cost, is in the η battery.

A Notation

symbol	meaning	value in the runs
a_0	mass of the component, in units of the background mass	0.0065–0.5 (blob: 10^{-3} –0.3)
L_0	common angular momentum of all particles	2
J, Q	radial action and angle; $Q = 0$ at pericentre	$J \in [0, 0.138]$
$\Omega(J)$	radial frequency of the orbit of action J	0.06–0.18
$\Omega_{\min}, \Omega_{\max}, \Delta\Omega$	edges and width of the frequency band	$\Delta\Omega = 0.011$ –0.050
$\tau_1 = 2\pi/\Delta\Omega$	time over which mixing dephases the $k = 1$ harmonic	125–560
J_t, g, W_0	edge of the support in action, exponent of the vacuum edge and depth of the Maxwellian	0.138; 2 (or 1); 3
$\varepsilon, s(J)$	amplitude and radial profile of the perturbation $\varepsilon s \cos Q$	$\varepsilon = 0.003$ –1; $s = (J/J_t)^{3/2}$
η	self-gravitational frequency shift over the width of the band	0.1–5.1 (blob: 0.005–2.5)
η_c	value of η at which the mode leaves the band	1.1–1.2
ω, γ	frequency and damping rate of the collective mode	$\omega = 0.061$ –0.105; $1/\gamma = 170$ –2700
$\mathcal{L}$	linearized Vlasov–Poisson operator, $i\partial_t\delta F = \mathcal{L}\delta F$; the frequencies of the discrete modes are its eigenvalues	–
x	position of the mode in the band, $(\omega - \Omega_{\min})/\Delta\Omega$	–0.59 (M5) to ~ 0.5 (A1)
ω_b	bounce frequency of the trapped particles	7×10^{-4} – 2×10^{-2}
$\nu = \omega_b/\gamma$	O’Neil parameter: trapping versus damping	0.17–10
$\mu = \omega_b/\Delta\Omega$	trapping versus the mixing of the whole band	≤ 0.37 (blob: ≤ 1.2)
$\nu_1 = \omega_b(1)/\gamma$	design factor, $\nu = \nu_1\sqrt{\varepsilon}$	0.17 (A1) to 10.9 (A4)
J_r	action of the resonance $\Omega(J_r) = \omega$, with $\Omega(J)$ continued outside the support if needed	0.124 (A4), 0.140 (L5), 0.22 (M5)
$\delta\Phi, \delta\Phi_\varepsilon, \ \delta\Phi\ $	potential perturbation; the same minus the Z run and divided by ε ; its rms value over $3 \leq r \leq 15$	–
h_1, h_k	first and k -th harmonic in Q of the distribution, with the test function $B(J)$	–
R	envelope ratio of $\ \delta\Phi\ $, PIC over linear	1 when the PIC follows linear theory
t_{nl}	first time at which the PIC departs from linear theory by 10 %	–
N_{rc}, N_{pc}, N	nodes in J , nodes in Q and number of particles	400, 25, 10^4

B Phase-space figures

Each figure shows the 10^4 particles at four times: the initial condition, $\tau_1/2$ (the shear begins), $2\tau_1$ (the mixing of the $k = 1$ harmonic has been completed twice) and the end. The top row is (r, p_r) and the bottom row the angle–action variables of the equilibrium. The colour is the weight perturbation of each particle (Section 2.5): red, mass added; blue, mass removed. Since the colour travels with the particle, the figures show how the flow transports the perturbation.

At $t = 0$ the perturbation $\cos Q$ occupies the columns of the grid in (Q, J) and, in (r, p_r) , forms a red crescent on the pericentre side (small r) and a blue one on the apocentre side: the component breathes. Each row of constant J then advances with its $\Omega(J)$. In (Q, J) the columns tilt and thin, and in (r, p_r) the crescent winds into an ever tighter spiral. This is phase mixing. At the end the pattern is finer than the resolution of the figure and looks like noise, but it is the

same perturbation, wound up. A row of J that stops being horizontal, or particles above the dotted line (the edge J_t of the support), is not mixing: it is a change of action, which only the self-consistent field produces. This is what distinguishes D3, D6 and D7.

B.1 Reference runs

The seven Z runs have the same equilibrium and initial positions as their D runs, without perturbation. The colour is only the label $\cos Q_0$. The pattern winds up as in the D runs because the particles rotate with their frequencies, but the density does not change, and the rows of J stay horizontal (Section 5.2).

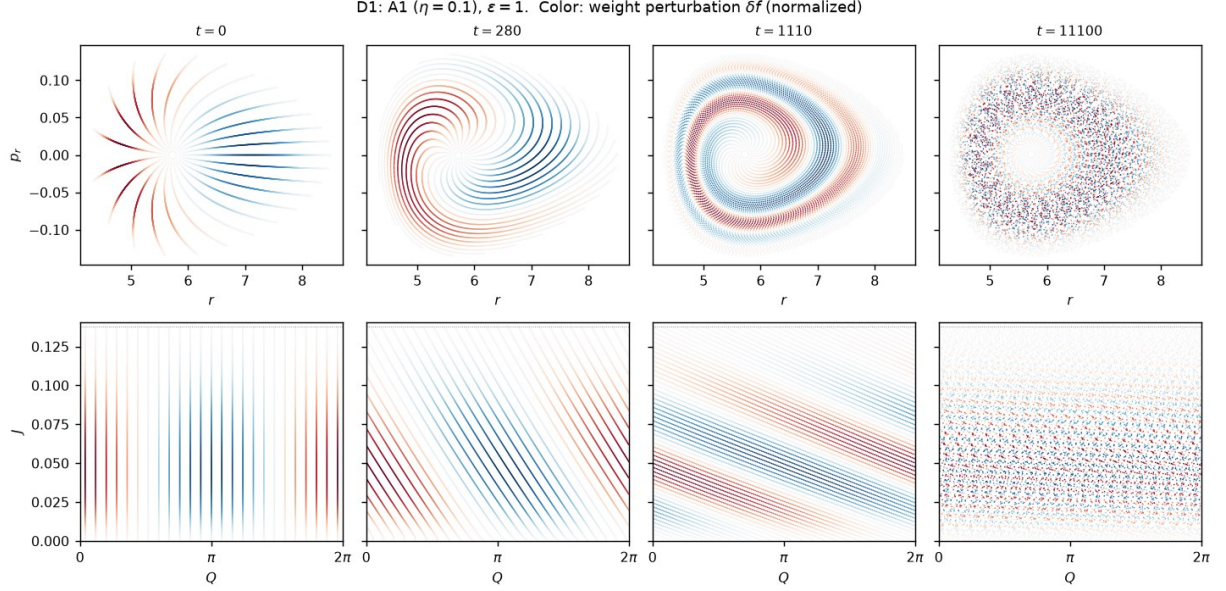


Figure 14: D1: $\eta = 0.1$, $\varepsilon = 1$. Almost pure free mixing: the rows of J stay horizontal; the maximum change of action is 0.002.

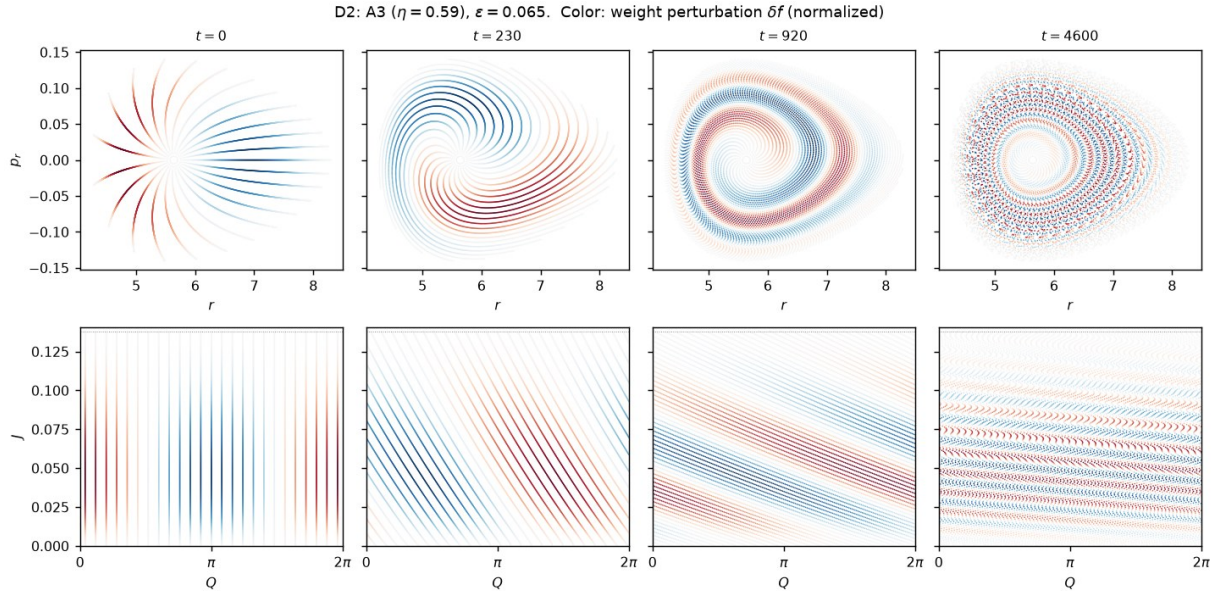


Figure 15: D2: $\eta = 0.59$, $\varepsilon = 0.065$. As D1, with a perturbation fifteen times smaller. The collective mode is damped more slowly, but this is not visible in (Q, J) because the amplitude is small.

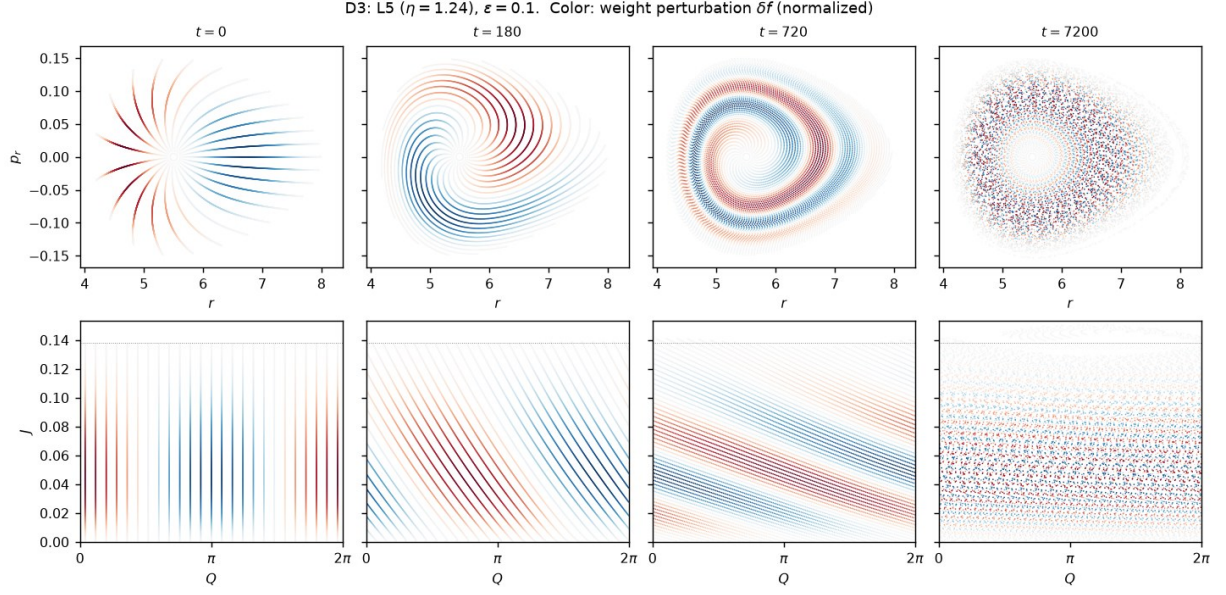


Figure 16: D3: $\eta = 1.24$, $\varepsilon = 0.1$, discrete mode just below the edge. In (Q, J) the edge rows ($J \approx 0.13$) detach outwards from the support; their action changes by up to 0.02 at the end (enlarged in Figure 7).

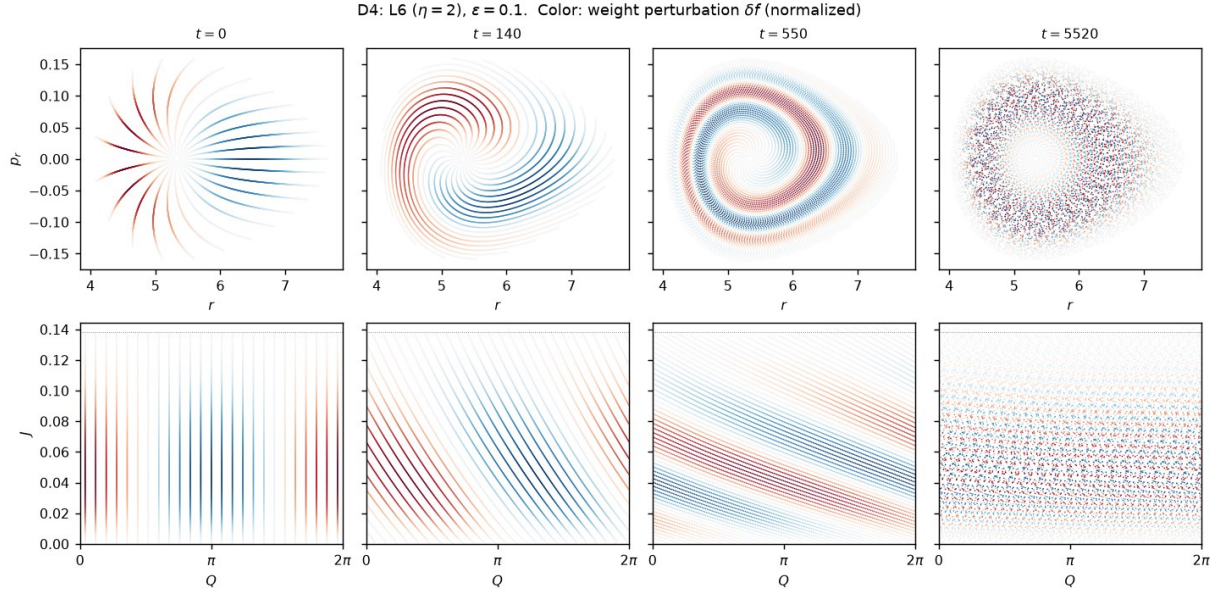


Figure 17: D4: $\eta = 2$, $\varepsilon = 0.1$, discrete mode separated from the band. The rows stay horizontal (maximum change of action 0.004).

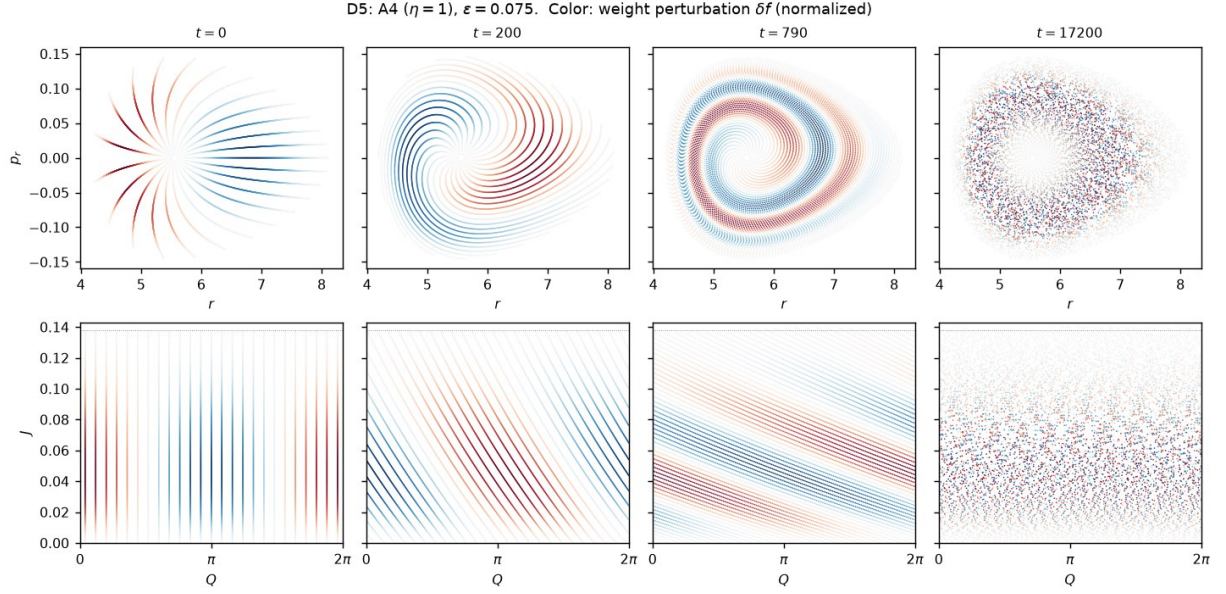


Figure 18: D5: $\eta = 1$, $\varepsilon = 0.075$, up to $t = 17200$ ($43 \tau_1$). The change of action is concentrated near the edge and saturates at 0.006: too small to be seen at this scale, but about fifteen times that of the unperturbed run.

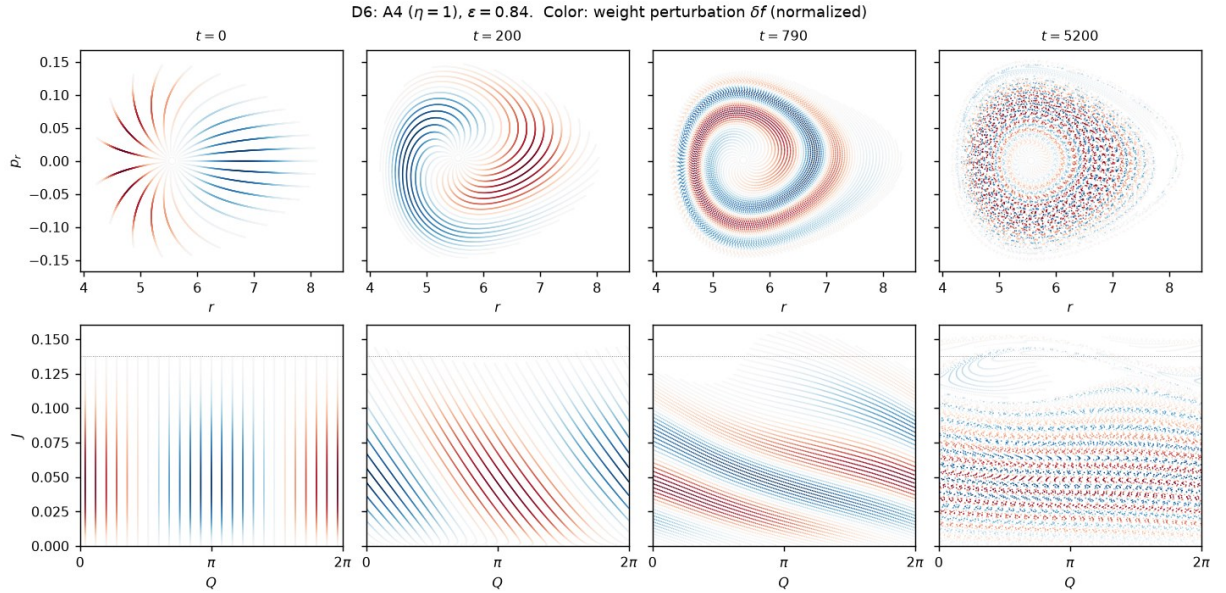


Figure 19: D6: $\eta = 1$, $\varepsilon = 0.84$. At the end ($t = 5200$) the rows with $J \gtrsim 0.11$ wind around a point: the island of particles trapped by the wave, next to the resonance $\Omega(J) = \omega$ near the edge. The change of action reaches 0.048.

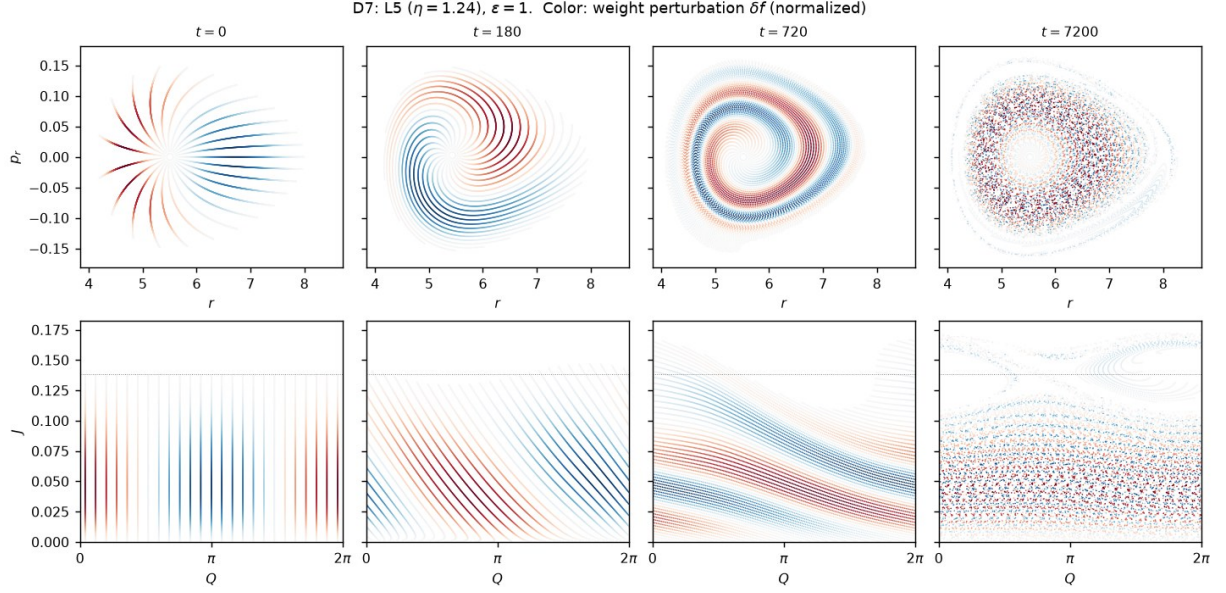


Figure 20: D7: $\eta = 1.24$, $\varepsilon = 1$. Already at $t = 720$ the rows undulate, and at the end there is an island next to the edge, with particles up to $J = 0.179$, outside the support. The change of action reaches 0.073, more than half the support.

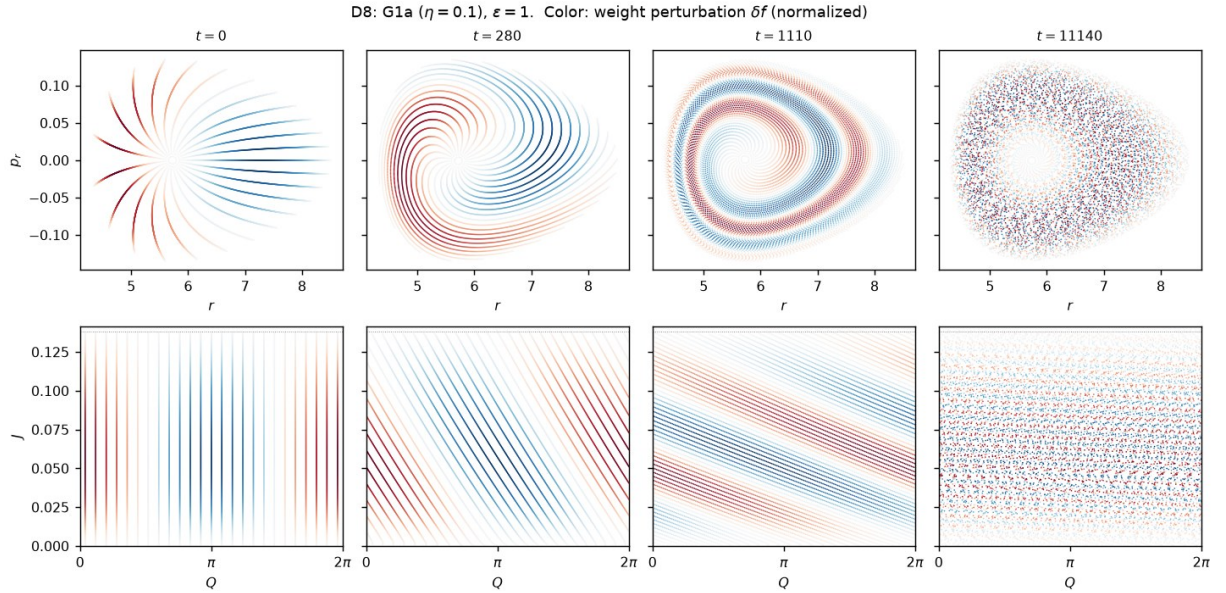


Figure 21: D8: $\eta = 0.1$, $\varepsilon = 1$, sharp edge ($g = 1$). As D1: the difference between the edges is not visible in phase space, because the persistent edge response is 10^{-3} of the initial value.

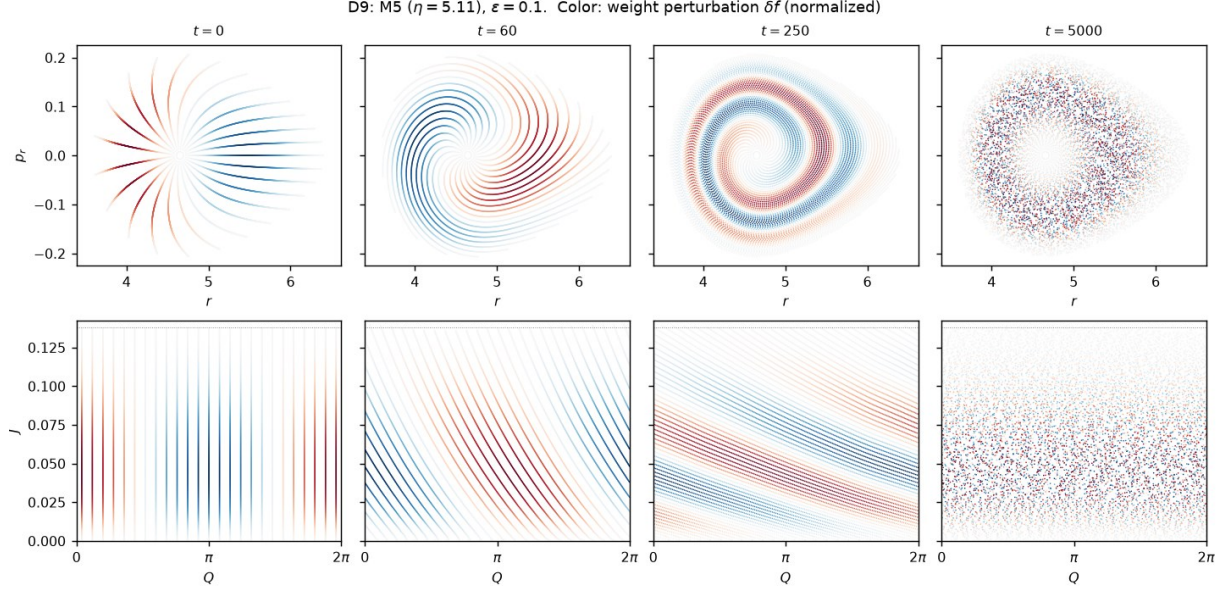


Figure 22: D9: $a_0 = 0.5$ ($\eta = 5.1$), $\varepsilon = 0.1$. The component is more compact ($r \in [3.6, 6.4]$) and the orbits are faster. The particles mix freely although the potential keeps oscillating without damping: the rows of J stay horizontal (maximum change of action 0.003).

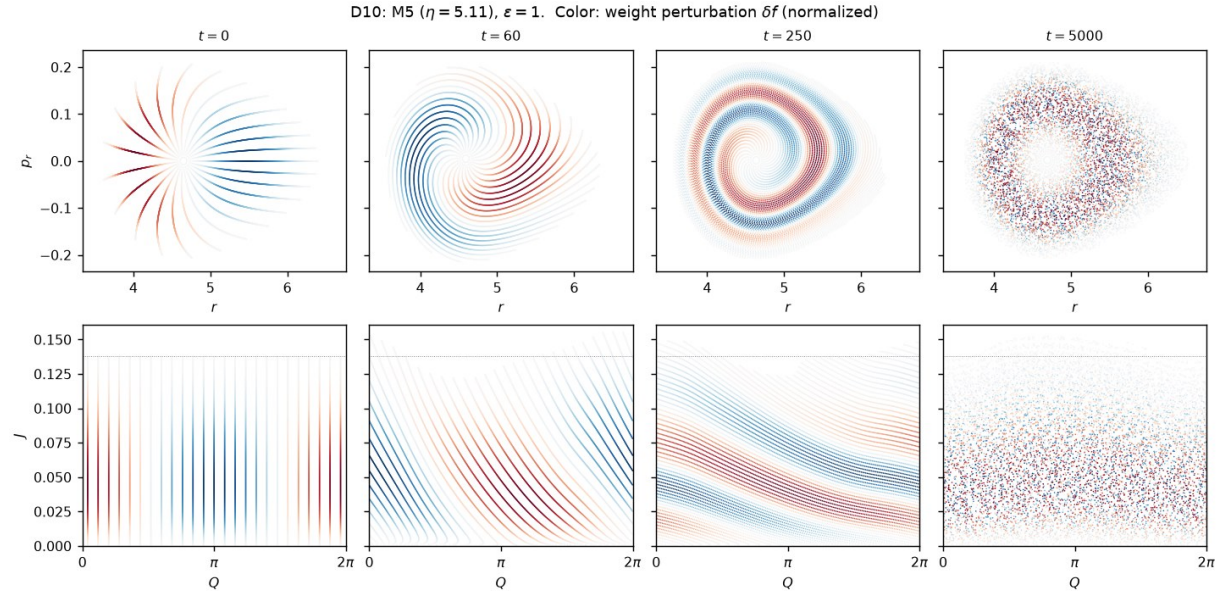


Figure 23: D10: $a_0 = 0.5$, $\varepsilon = 1$. The rows undulate slightly at $t = 250$ and some particles cross the edge J_t , up to $J = 0.157$, but no island forms: the resonance of the mode is at $J \approx 0.22$.

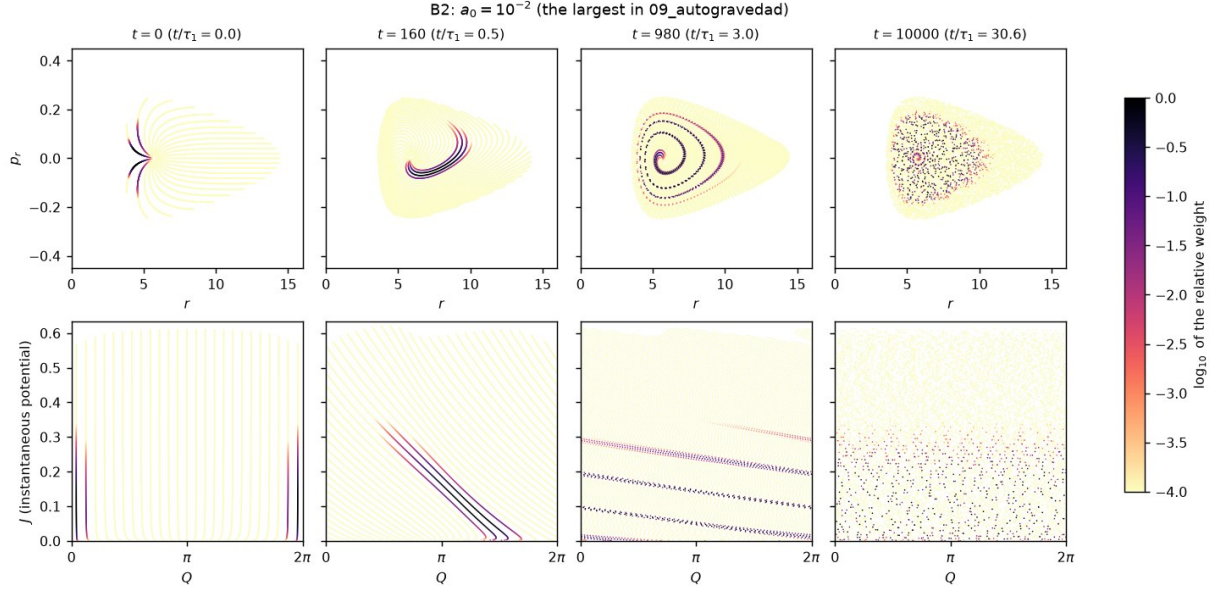


Figure 24: B2: the blob with $a_0 = 10^{-2}$, the largest mass of the earlier runs. Colour: logarithm of the weight of each particle (dark where the mass is). It is sheared into a spiral and mixes, but a remnant core remains at $r \approx 5.7$ at the end. Bottom: (Q, J) in the potential of each instant.

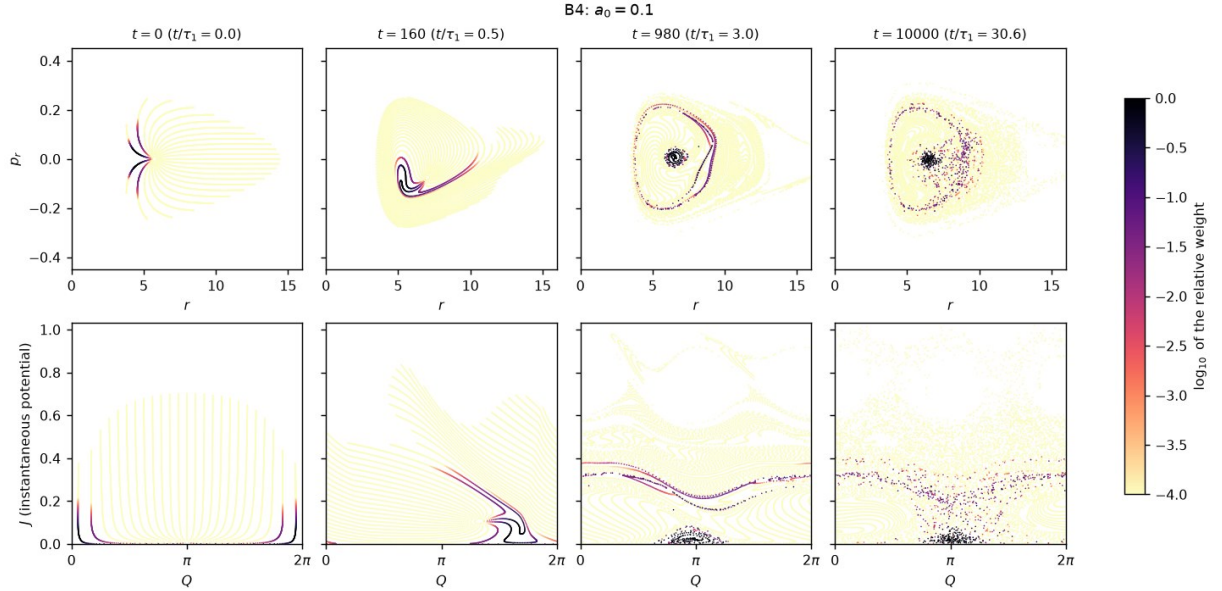


Figure 25: B4: the blob with $a_0 = 0.1$. It collapses in the first radial periods into a dense core ($r \approx 6.5$, $p_r \approx 0$ at these times) with 79 % of the mass, surrounded by a halo of stripped particles. In (Q, J) the core is at $J \approx 0$: it is the nearly circular orbit of the potential that it creates.

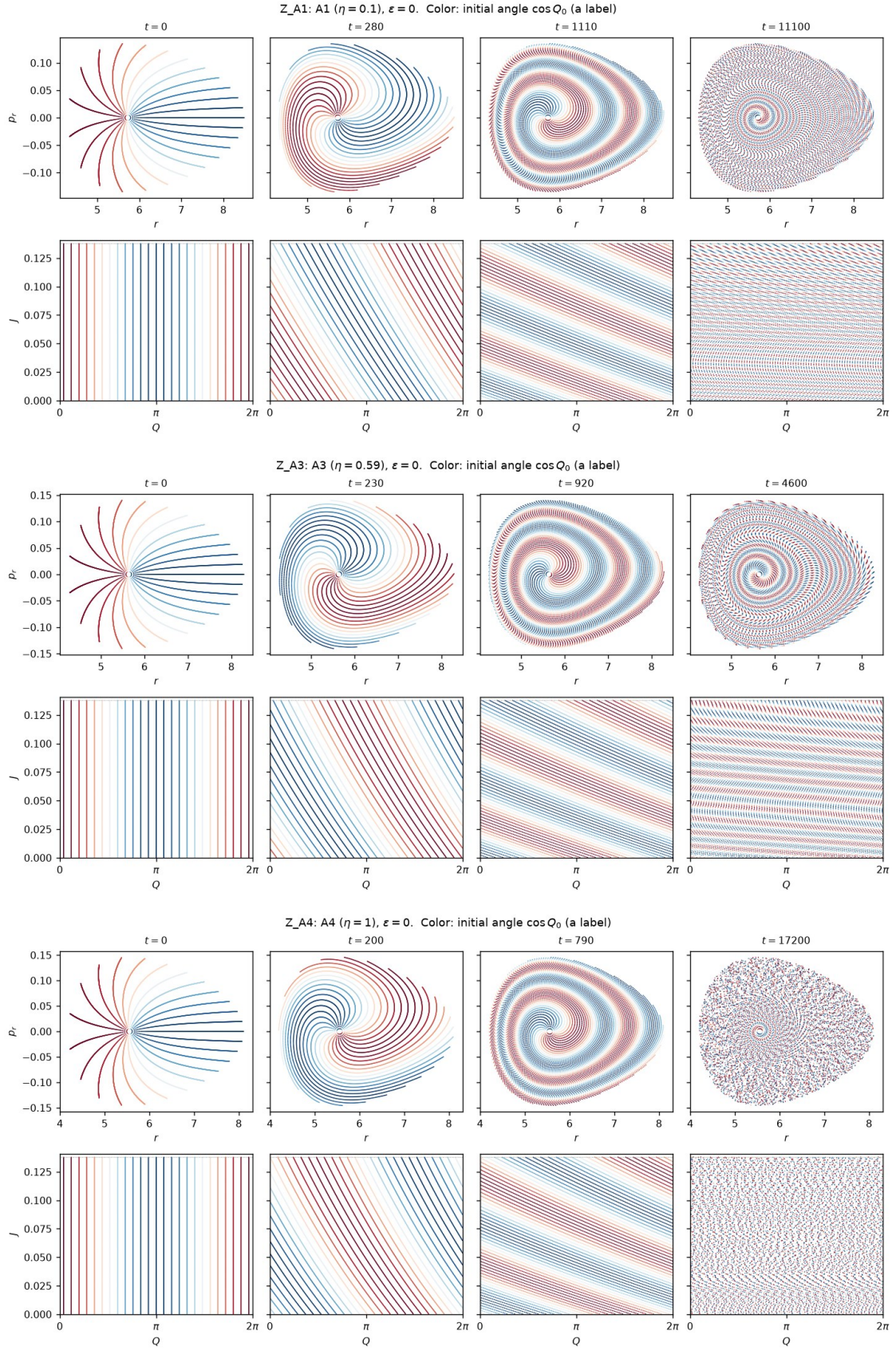


Figure 26: Reference runs of A1 (for D1), A3 (D2) and A4 (D5 and D6).

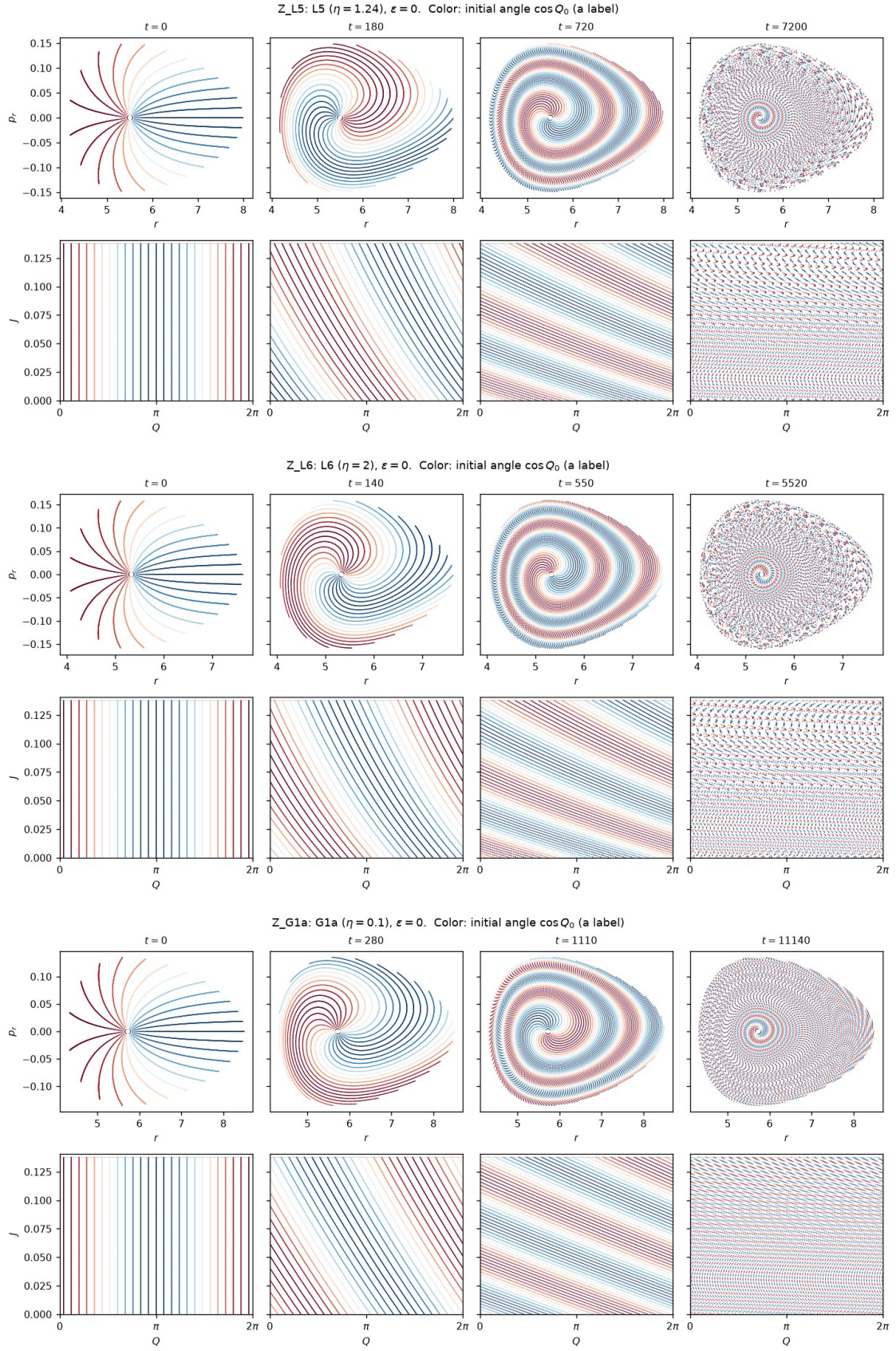


Figure 27: Reference runs of L5 (for D3 and D7), L6 (D4) and G1a (D8).

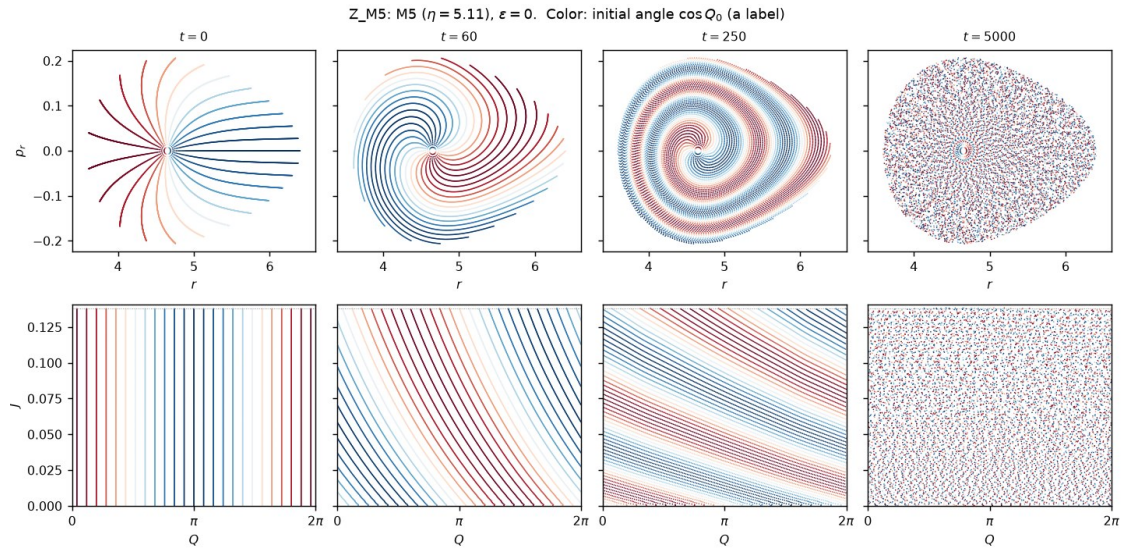


Figure 28: Reference run of M5 (for D9 and D10).

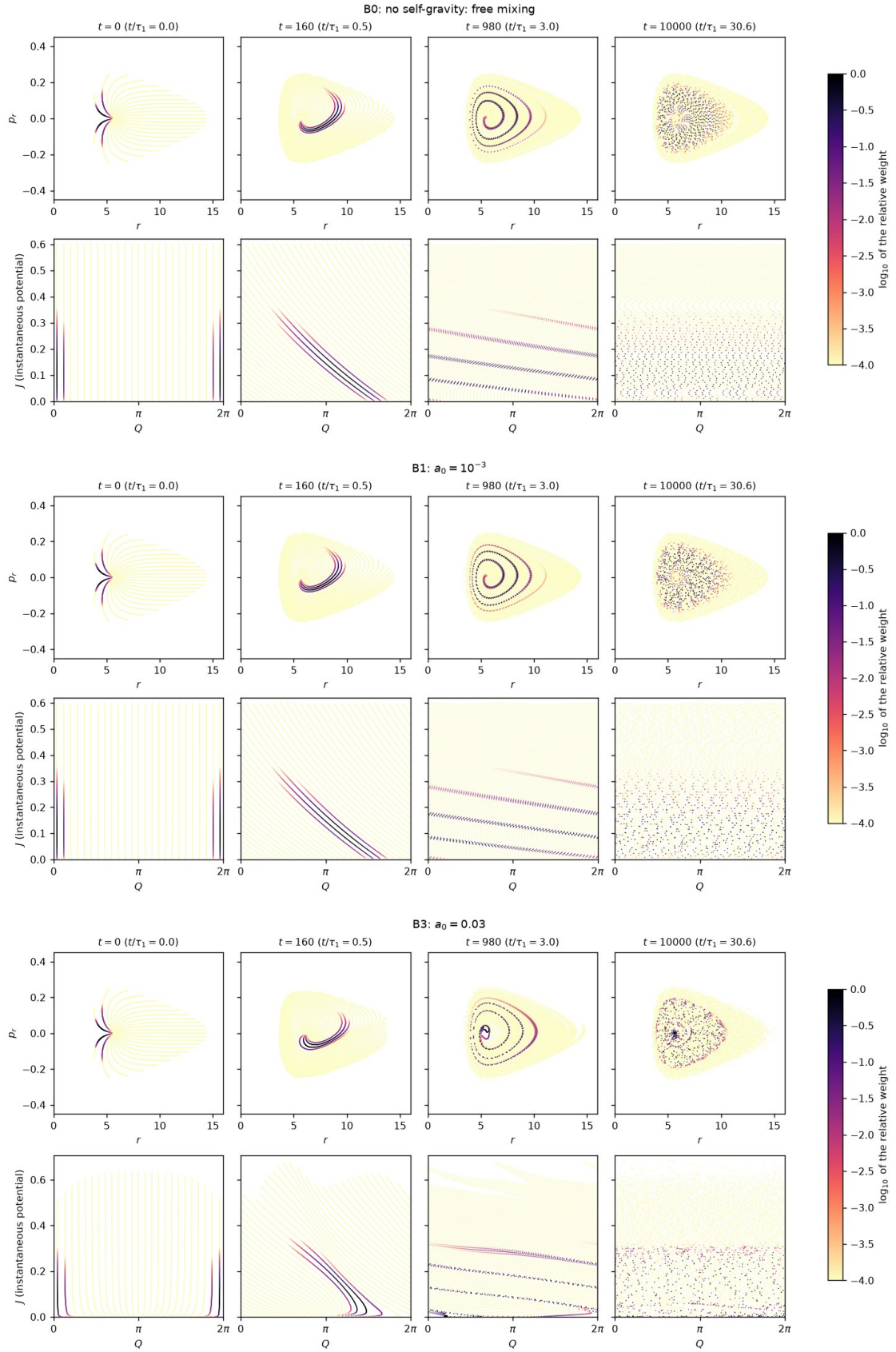


Figure 29: The blob without self-gravity (B0), with $a_0 = 10^{-3}$ (B1) and with $a_0 = 0.03$ (B3).

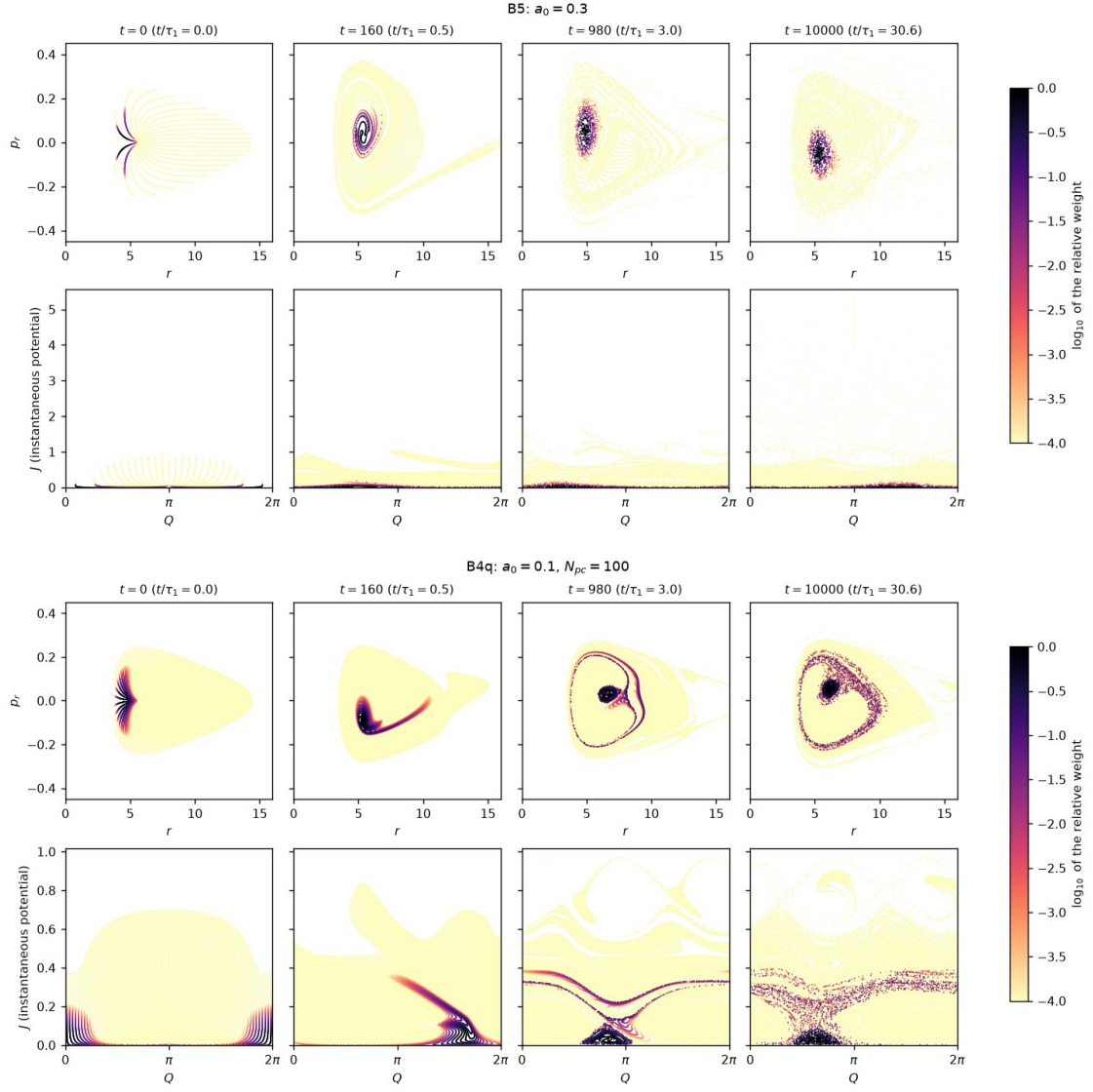


Figure 30: The blob with $a_0 = 0.3$ (B5) and with $a_0 = 0.1$ and $N_{pc} = 100$ (B4q).

C Videos

There is one video per simulation, twenty-four in total (the seven of the blob are named `bulto_B*.mp4`), in `reproducir/videos/demo_eta/<run>.mp4`. They are not under version control and are regenerated with `demo_eta.py videos`; the current files still have Spanish labels. They are 1280×720 pixels, with one frame per snapshot (every 10 time units, about eight per radial period) at 30 frames per second, and last from 15 s (D2) to 57 s (D5 and Z_A4). Each frame has four panels:

- top left, the particles in (r, p_r) , coloured by their weight perturbation as in Appendix B, which shows the mixing;
- top right, the particles in (Q, J) , coloured by their *initial* action as in Figure 7, which shows changes of action: under free mixing each colour stays in its row, and a row that deforms, or crosses the edge J_t (dotted line), has changed its action through the self-consistent field;
- bottom left, the profile $\delta\Phi_\varepsilon(r)$ of the PIC run and of linear theory at that time, with the vertical axis rescaled to the envelope so that the profile remains visible as it decays;
- bottom right, the envelope of $\|\delta\Phi\|$ over the whole run, PIC and linear, with a line marking the time of the frame.

In the videos of the Z runs the colour in (r, p_r) is the label $\cos Q_0$, and the bottom panels show the $\delta\Phi$ of the unperturbed run relative to the equilibrium potential. Its static part is the error of the PIC field. In Z_L5, Z_L6 and Z_M5 it also oscillates at the frequency of the discrete mode, which that error excites (Section 5.2); in the other Z runs the oscillation is the noise floor.

In D1 and D2 the spiral tightens while the profile of $\delta\Phi$ dies out. In D3 and D4 the profile still oscillates in the same way at $20\tau_1$, although the particles have mixed: the mode is collective, not a group of particles. D6 and D7 show the formation of the island in (Q, J) . At the end of D5 the PIC curve separates from the linear one and stops decreasing; with four times more particles the separation is much smaller, so much of it is discreteness noise (Section 5.3). D10 contrasts with D7: at the same amplitude, but with the mode far from the band, the rows in (Q, J) move little, no island forms, and the profile of $\delta\Phi$ follows the linear one.

D External review

An external review of this work (`auditoria_deepseekpro/revision_cientifica.md`) identified what was missing to turn the demo into a result. The point-by-point response is in `auditoria_deepseekpro/respuesta_revision_cientifica.md`. In summary:

- **Addressed in this version:** the uncertainties of the poles (Table 6); the accuracy of the fit, tested with synthetic signals; energy conservation; the smooth/fine split at three scales (Section 5); explicit predictions; the limitations of the fixed- L model; and conclusions separated into demonstrated, interpreted and hypotheses.
- **Done with new runs:** the ε scan of D3, the convergence in N and Δt of D3, D5 and D6, and the comparison of the island of D6 with the pendulum (with D6L, up to $43\tau_1$). D3 and D6 are converged; D5 at late times is not.
- **Discussed with the data:** whether the loss of D3 depends on the fit (the linear envelope bounds it without a fit); whether the Q factor in the code's h_k convolves harmonics (the code multiplies each harmonic by a constant); the half-width of the pendulum island, which is $2\sqrt{A/|\Omega'|}$; whether the fine part should decrease as $N^{-1/2}$ (with a quiet start the noise is not Monte Carlo noise); and whether the comparison with Hadžić et al. is only an analogy (their theorem is formulated in this same fixed- $|L|$ reduction, Section 1.6).

- **Pending:** D5 with $N \times 16$; the robustness of η_c with respect to the edge (including $g = 1/2$), the width and W_0 in the linear solver; D8 at the optimal amplitude; $\eta_c(L_0)$; the quantitative comparison with O’Neil’s formulas; and the point-mass case of the theorem.

E Reproduction

The whole workflow is in `reproducir/scripts/demo_eta.py`. It has six steps, which are run from that directory and reuse whatever already exists:

step	what it does	time
preparar	the 17 initial states, with <code>equilibrio.py</code> , and their <code>.par</code> files in <code>reproducir/corridas/12_demo_eta/</code>	4 min
lineal	the 7 linear solutions with <code>lineal.py</code> , up to the longest t_{fin} on each equilibrium	15 min
correr	the 17 PIC runs, one after another, with 4 threads	3.3 h
analizar	angle–action map of each snapshot (computed once), h_k and $\delta\Phi$; 4 processes	55 min
figuras	phase-space figures, time series and <code>exe/demo_eta/resumen.txt</code>	2 min
videos	one video per run, 4 at a time	50 min

The steps `controles`, `isla`, `energia` and `resolucion` produce the tables and figures of Section 5, and `armonicos` the harmonic content of the discrete modes (Section 1.5). The two constructions of the equilibria are compared by `comparar_equilibrios.py` (Table 2; with `--refinar`, also the finer grids of the iteration). Their evolution with the PIC code is run by `equilibrio_edo_pic.py`, with the steps `preparar`, `correr` (8 runs, 1.7 h) and `analizar`. The last one also reads the Z runs analysed by `demo_eta.py`, and writes the numbers of Section 5.2 to `exe/demo_eta/edo/resumen.txt`. The blob has its own script, `reproducir/scripts/bulto.py`, with the steps `preparar`, `correr` (7 runs, 1.6 h; B4q alone, 47 min), `analizar` (1.1 h: each snapshot is mapped in the instantaneous potential and in the bare isochrone), `figuras` and `videos`. Its outputs are in `exe/demo_eta/bulto/`. All outputs take 2.6 GB in `exe/demo_eta/`, outside version control.